Growth and Labor Reallocation: Vertical versus Horizontal Innovation *†

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Abstract

Labor reallocation is often regarded as an essential component of productivity growth, based on the thought that innovation is made by firms expanding to new business areas with better technology (horizontal innovation). I study the relationship between long-run growth and labor reallocation by introducing innovation made by incumbent firms previously operating in the product market (vertical innovation). I use novel data from a German establishment-level panel survey to distinguish innovation types from different survey questions about innovation activities and construct an endogenous growth model with different types of innovation to decompose the relative contribution of innovation types on long-run growth. I calibrate the model with the survey data merged with the administrative employer-employee matched data and use the calibrated model to quantify the aggregate effect of the Hartz reform (which increased labor supply and reduced labor market rigidity in Germany during the mid-2000s). I find out that vertical innovation that does not cause direct creative destruction accounts for more than 90% of productivity growth on the balanced growth path. Moreover, vertical innovation generates 85% of dynamic gains from faster productivity growth after the reform. Since vertical innovation has a smaller effect of reallocating labor inputs toward innovating establishments, despite the sharp increase in output growth, labor reallocation rates, especially job destruction rates, do not increase but decrease over the transition path.

Keywords: Innovation, Creative Destruction, Productivity Growth, Labor Reallocation
JEL codes: O33, O40, E24

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†This study uses the LIAB longitudinal model, version 1975-2017, of the Linked-Employer-Employee Data (LIAB) from the IAB. Data access was provided via on-site use at the Research Data Centre (FDZ) of the German Federal Employment Agency (BA) at the Institute for Employment Research (IAB) and subsequently remote data access. DOI: 10.5164/IAB.LIABLM7517.de.en.v1.

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1 Introduction

The rates at which labor inputs are reallocated across firms - job creation rates for growing firms and job destruction rates for shrinking firms - have been an informative metric to evaluate the growth potential of the economy. Behind this is the Schumpeterian thought that labor reallocation from low to high productivity firms is an essential process for productivity growth. Therefore, most firm-level innovation models assume that for a product to be improved, there must be a new firm that replaces the current firm. A seminal paper by Aghion and Howitt (1992) models innovation as firms competing against each other to occupy a limited number of products. In this model, it is always the intruders that make innovation and drive out the current top firm with cutting-edge technology. Most firm-level innovation papers follow this tradition and therefore model innovation promoting competition across firms[1].

In this paper, I study how growth is correlated with the labor reallocation process when both current and expanding producers can improve the quality of goods. If current producers innovate the goods they previously made, the old jobs of the producers are not destroyed by new producers; the amount of labor reallocation made by this type of innovation is smaller than the innovation replacing old producers with new producers. Thus, how much labor reallocation is needed for a given level of productivity growth depends crucially on which type of innovation is dominant in the economy.

Throughout this paper, I define three types of establishment-level innovations: vertical, horizontal, and entrant innovations. First, vertical innovation is defined as improvements of quality or production efficiency by a firm currently operating in the product market. The German firm, 'Bosch', for example, started to make power drills in 1932 and has continuously made quality improvements on the same item until now (See the left panel of Figure 1.).

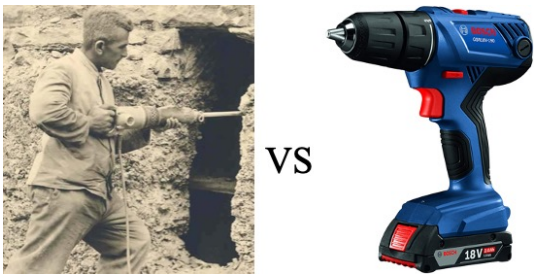
[1]For example, see Klette and Kortum (2004), Lentz and Mortensen (2008), and Acemoglu et al. (2018) among many.

The newer version of drills by Bosch replaces the older version of Bosch drills. Thus, this type of innovation increases the quality of a product without firm replacements[2].

Another type of innovation in this paper is horizontal innovation, defined as an expansion of a firm new to the product market. In the right panel of Figure 1, the same German Firm, Bosch, started to offer electric motors for the e-bike in the early 2010s. This expansion would put Bosch in competition with other firms already producing electronic motors for e-bikes. If the expanding firm, Bosch, is more productive or offers a better quality good, it hires new workers and replaces other firms. The other firms reduce their labor inputs. Thus, because of horizontal innovation, net labor reallocation takes place at a much larger magnitude. Lastly, entrant innovation is made by entrant firms. When the number of markets is limited, entrant firms also replace existing firms in markets.

Figure 1: Innovation Examples of Bosch

(a) Vertical Innovation



Generations of Hand Drills by Bosch

(b) Horizontal Innovation

Bosch's expansion to E-Bike Motors

Despite the important differences between vertical and horizontal innovations, no unified framework has been made to contain both types of innovations[3]. This is because measur-

[2]All images of Figure 1, as well as the history of the company, are from the website of Bosch, `https://www.bosch.com/company/our-figures/`.

[3]A few exceptions include Garcia-Macia et al. (2019) and Akcigit and Kerr (2018). Without a measure of innovation, Garcia-Macia et al. (2019) build a parsimonious model with fixed innovation rates and estimate underlying innovation parameters using establishment-level employment data. Akcigit and Kerr (2018) use a citation network of patents to distinguish innovations within firms' existing technology clusters and from innovations in new areas.

ing vertical and horizontal innovations separately has been difficult. Thus, economic papers often assume that innovation activities measured in the data are expansion to product markets new to the firm as opposed to improvements in the previously occupied markets. This assumption is based on the Arrow replacement effect: that a monopolistic firm has smaller incentives to innovate compared to an outside firm because the new product replaces the old product supplied by the monopolist.

I show the relative contribution of innovation on productivity growth and labor reallocation using a quantitative model calibrated from empirical moments about innovation activities. I use novel establishment-level survey data from Germany to identify both the innovation on firms' existing products and the innovation to expand to new business areas. I distinguish these innovations separately from the survey questions that directly ask whether each establishment had improved the quality of their products, upgraded production processes, and added a new product variety into their portfolios. I merge the survey data to the administrative employer-employee matched data where I can observe the number of employees at each establishment before and after the innovation events. The employment growth after innovation activities informs the model about how much productivity gains are associated with the surveyed innovation measures.

Then, I construct an endogenous growth model with innovation choices to translate individual innovation activities into aggregate outcomes on productivity growth and labor reallocation. The model parameters are calibrated to match empirical moments from the survey and the administrative data. Then, the model is used to quantify the relative contribution of different innovation types. Lastly, I use the model to evaluate the impact of a major labor market in Germany on aggregate outcomes. The reform increases labor supply and demand exogenously and becomes a useful structural break to see the relationship between growth and labor reallocation.

On the empirical side, I establish new findings from the survey-based measures of

innovation. I find out that vertical innovation happens more frequently (30 percent in a year) than horizontal innovation (20 percent in a year). Moreover, horizontal innovation is more strongly associated with labor reallocation measures than vertical innovation. Both innovation types predict a 1 percentage point increase in employment growth rates at the establishment level one year after the innovation. However, while the correlation of establishment-level employment with horizontal innovation is accumulated and becomes even larger (3 percentage points) and more significant four years after the innovation, vertical innovation is not positively correlated with employment growth after four years without further innovation. Moreover, at the sector level, the fraction of horizontally innovating establishments is significantly negatively associated with the size growth of non-innovating establishments. The coefficient of vertical innovation frequency is insignificant and much smaller in magnitude.

The model in this paper encapsulates these facts in a quality ladder growth model with two types of incumbent innovation. Vertical innovation is modeled as improvement of the quality by the firm currently occupying the product line. This innovation leads to within-establishment productivity growth that has limited impacts on job flows across firms. On the other hand, horizontal innovation is an improvement of the quality by the firm expanding to a new product line. This innovation leads to direct creative destruction with the expanding firm replacing an old firm on the product line. It is shown that job creation and destruction are larger in magnitude after horizontal innovation than vertical innovation in the model.

The model, after being calibrated to match the innovation and employment growth moments, highlights that the vast majority of productivity growth comes from vertical innovation. More than 90% of productivity growth on the Balanced Growth Path (BGP) comes from vertical growth while horizontal and entrant innovations account for only 3% and 6%, respectively. Meanwhile, for job creation, vertical innovation accounts for only a quarter

of new jobs. It is entrant innovation that accounts for half of the new jobs on the BGP. Horizontal innovation makes up the remaining quarter of job creation.

Then, I use the quantitative model to evaluate macroeconomic effects from a major labor market reform in Germany. 17% of total welfare gains from the reform come from faster productivity growth through more vertical, horizontal, and entrant innovation. Among these margins, vertical innovation accounts for 85% of the productivity growth gains. Since vertical innovation does not generate much job destruction, job destruction rates decrease over the transition path after the reform despite increases in output growth. This model prediction is consistent with the declines in average job destruction rates after the Hartz reform in the data.

Related Literature

This paper contributes to the field of the endogenous growth model by extending the classic quality ladder model à la Aghion and Howitt (1992) and Klette and Kortum (2004) with vertical innovation choices and product-line heterogeneity. Following these papers, endogenous growth models have focused on horizontal innovation that replaces firms operating on the product lines with more productive new firms. Lentz and Mortensen (2008) introduce type heterogeneity of firms and productivity heterogeneity across product lines to match moments about the firm size and firm growth rates. More recently, these models are used to address various issues such as the age-dependency of innovation intensities (Acemoglu et al., 2018), the cross-country difference in expansion capacity and productivity growth (Akcigit et al., 2021), and the role of horizontal innovation to reduce mark-ups and misallocation in the market (Peters, 2020). This paper incorporates vertical innovation of incumbent firms currently operating on the product lines, as in Akcigit and Kerr (2018), Acemoglu and Cao (2015), and Garcia-Macia et al. (2019). This paper finds that vertical innovation is more important than horizontal innovation for productivity growth. Additionally, this paper in-

troduces endogenous innovation choices to Garcia-Macia et al. (2019) and heterogeneous returns to horizontal innovation from productivity differences over product lines to Akcigit and Kerr (2018). The model in this paper derives welfare implications of policy changes through an endogenous innovation margin as in Atkeson and Burstein (2019). In the model, increasing labor supply and removing firing costs improve welfare through more vertical innovation[4].

Second, this paper contributes to discussions on the aggregate effect of innovation on employment. Papers on this issue often rely heavily on the underlying model to back out the importance of different types of innovation. Pissarides and Vallanti (2007) start from the theoretical observation that growth is positively correlated to unemployment if new technology is embodied in new jobs[5]. Given that growth is mostly negatively correlated with unemployment in the cross-section of industrialized countries, Pissarides and Vallanti (2007) conclude that embodied technology plays no role in shaping the steady-state patterns of unemployment. Postel-Vinay (2002) complements this idea by underscoring that the short-run response to technological improvements embodied in new jobs is an increase in employment, unlike the long-run response (due to forward-looking behaviors in the job creation side). Hornstein et al. (2007) further apply this idea to show that embodied technology can explain key labor market differences between the U.S and Europe. Using high-quality micro-data on establishment-level employment within the United States, Garcia-Macia et al. (2019) discover that the contribution of innovation associated with creative destruction must be minor because most job creation and destruction at the establishment level are small in magnitude than what the creative destruction model of labor adjustment implies. In the

[4]In Appendix C.3, I derive growth and welfare implications of R&D tax incentives. When only vertical innovation is subsidized, welfare, as well as productivity growth, improve. However, when only horizontal innovation is subsidized, despite productivity growth gains, the welfare effect is negative because horizontal innovation consumes too much production labor.

[5]Michau (2013), however, shows that unemployment does not necessarily increase with creative destruction if on-the-job search directly reallocates workers toward high productivity jobs.

declining business dynamism literature, Decker et al. (2014, 2016) discuss that the lower degree of entrant innovation in the U.S. is the key factor for slower labor reallocation in the U.S. Using a more direct measure of innovation, the findings in this paper suggest that innovation does not increase unemployment quantitatively because most innovation does not directly destroy jobs at old firms.

Third, this paper contributes to the literature on empirical measures of establishment-level innovation. Establishment-level innovation behavior is difficult to measure; economists often rely on indirect inference methods to back out innovation behavior or productivity shocks from other establishment-level behaviors such as employment, output, and wage as in Lentz and Mortensen (2008), Coles and Mortensen (2016), and Garcia-Macia et al. (2019). On the other side, many studies use the patent application, grant, or citation data to measure innovation activities of a sector, an industry, or a firm. For instance, Aghion et al. (2005) use the number of patents granted in the United Kingdom to show that innovation is overall inverse-U shaped in the degree of competition at the sector level. More recently, Dorn et al. (2020) matches patent assignees in the United States Patent and Trademark Office to public firms in Compustat and states that firms more subject to import competition produce fewer patents. Akcigit and Kerr (2018) combine patent data with Census data to inform how external and internal innovation activities proxied by patenting activities are associated with firm growth. Moreover, Kogan et al. (2017) measure the value of patents using stock market prices of firms before and after the day of patent grant[6]. This paper uses new survey-based measures that capture innovation behaviors of establishments far from the technology frontier across different sectors, including service sectors[7].

Lastly, this paper is related to the literature that evaluates the quantitative impact

[6]For more examples, see Aghion et al. (2005, 2009); Hashmi (2013); Seru (2014); Webb (2019); Argente et al. (2020).

[7]Survey-based measures of innovation are recently getting attention in economics. Zolas et al. (2021) propose to use a new module on innovation activities in the 2018 Annual Business Survey to measure advanced technology adoption such as cloud computing and artificial intelligence.

of the Hartz labor market reform. Because of its scope and depth in restructuring public employment agencies, the unemployment benefit system, and employee protection, the Hartz reform has been considered one of the most important changes in the labor market system in post-war Germany. However, most papers focus on the static impact of the reform on employment, wage, and welfare. Krause and Uhlig (2012) study the effectiveness of the Hartz reform using a search and matching model of the labor market. They conclude that the Hartz reform reduced the steady-state unemployment rate by 2.8 percentage points. More recent studies such as Launov and Wälde (2013), Launov and Wälde (2016), Bradley and Kügler (2019), and Hochmuth et al. (2021) also model frictional labor markets to quantitatively assess the effect of the reform. This paper considers the innovation choice of establishments to be another mechanism through which the reform affects the welfare. The labor market reform affects R&D costs and future returns to innovation by increasing labor supply and removing labor market rigidity. Then, an increase in innovation may improve the welfare of the representative household in a world with positive externalities of innovation.

This paper is organized as follows. In Section 2, I describe the German establishment panel survey and a motivating relationship between incumbent innovation and firm behavior. In Section 3, I build a quantitative model of endogenous incumbent innovation choices in the general equilibrium setting. In Section 4, I explain how model parameters are identified from empirical moments in the German data. Lastly in Section 5, I use the calibrated model to quantitatively evaluate the impact of the Hartz reform through the endogenous incumbent innovation margin.

2 Evidence on Innovation and Labor Reallocation

2.1 Data Description

I use German establishment panel data to measure the two innovation activities directly. The data is from the IAB Establishment Panel (IAB-BP), a representative establishment survey of the Research Data Centre (FDZ) in Germany. The survey contains around 17,000 observations per year and is implemented annually from 1993 for Western Germany and 1996 for Eastern Germany. The data contains a wide range of establishment-level information about employment, hiring, separation, wage, business volume, investment, and innovation activities. It is possible to merge the survey to the administrative employer-employee matched data set - the Linked Employer-Employee Data from the IAB (LIAB) - using the common establishment identifier. In order to avoid possible measurement errors, I focus on the number of employees as of June 30 in the administrative data set as my measure of establishment size.

The notion of establishment in the German data is modestly different from the U.S. Census Bureau's concept of establishment. In the U.S., establishments are distinguished by their physical location. In Germany, however, a common establishment code is shared by multiple workplaces at different locations if they belong to the same company, are physically located in the same municipality, and have the same economic class. Still, a company may hold different establishment codes across different municipalities and economic classes.

Common challenges with establishment survey data are: under-representation of entering establishments and attrition of establishment samples that are not related to the actual establishment exits. For entrants, the survey reports a question on the year of foundation in the current form[8]. I calculate the establishment age from answers to this question. Es-

[8]The actual question is 'Is this establishment/office founded before/after 1990?' If the answer to the previous question is 'after 1990', then they are further asked 'Please indicate the year the establishment was founded'.

tablishments which were founded either in the survey year or the previous year are defined as entrants. For exiting establishments, I identify exiting establishments from the variable indicating the last year the establishment code exists in the administrative data. Because of potential selection issues associated with establishments that exit the market and continuing establishments that simply disappear from the panel, I focus on the establishments that survive in the survey in the baseline case.

The two measures of innovation activities are derived from the answers to the following questions in the survey[9].

1. **Horizontal Innovation**: In the last business year, did your establishment start to offer a product/service that had been available on the market before?

2. **Vertical Innovation**: In the last business year, did your establishment improve or further develop a product or service which had previously been part of your portfolio (Quality Improvement)? Or, did you develop or implement procedures in the last business year of 2015 which have noticeably improved production processes or services (Process Innovation)?

These innovation variables are surveyed consistently starting from 2008; because of this, I use observations from 2007 (2008 survey) to 2017 (2018 survey). All establishment-level observations on innovation activities are weighted by the cross-sectional weight constructed by the data provider.

Figure 2 shows the fraction of establishments with each of the two innovation activities in each size group. All year × establishment observations are pooled in the sample[10]. Establishments

[9]The innovation module of IAB-BP has four distinct questions on the innovation activities of which the three are included. For the description of all the questions, see Appendix A.1.

[10]In Appendix A.1, I show the number of innovation years for each establishment. There exists a substantial heterogeneity; i.e. most establishments do not vertically or horizontally innovate at all while some establishments innovate every year. As a result, the fraction of vertically and horizontally innovating establishments before and after vertical and horizontal innovation events in year 0 is substantially higher than the average fraction of vertically and horizontally innovating establishments. These correlation statistics are

are grouped by size, and size is defined by the number of employees measured two years before the survey[11].

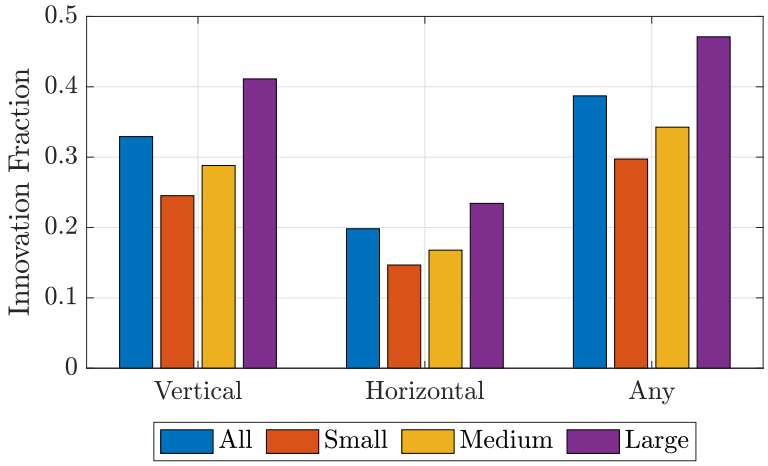
Overall, more establishments are engaged in vertical innovation than horizontal innovation. However, these two types of innovation are partially synchronized since the fraction of establishments that experienced vertical innovation is close to the fraction of establishments that experienced either vertical or horizontal innovation, or both. In the descriptive analysis and the calibration exercise in the paper, I use unconditional moments on establishment growth after vertical and horizontal innovation.

In Figure 2, larger establishments are more likely to experience both vertical and horizontal innovation than smaller establishments. This size dependency will be used later in the calibration of the model. Establishment size is a (noisy) proxy for the underlying productivity of the establishment. Therefore, the size dependency of innovation frequencies informs the model of how elastically establishments are responding to the underlying differences in productivity when making an innovation decision.

also described in Appendix A.1.

[11] The survey asks about innovation activity in the previous year of the survey. As a result, the relevant predictor for the innovation frequency is the employment size as of June 30 two years before the survey. When the establishment does not have the information on the previous employment size, it is omitted from the samples used in Figure 2.

Figure 2: Innovation Frequencies across Size Groups(in %)



In Appendix A.2. I compare the survey-based measure of innovation to the traditionally used patent-based measures at the sector × year level. The number of patents per establishment in each sector is positively correlated with the fraction of either horizontally or vertically innovating establishments. The fraction of vertically innovating establishments is positively associated with the number of internal patents[12] that cite the previous patents of their assignees for more than half of the total citations. However, the fraction of horizontally innovating establishments is uncorrelated with the number of external patents that cite the previous patents of their assignees for less than half of the total citations.

2.2 Establishment-level Innovation and Growth

To infer the quantitative effect of innovation, I show how establishments grow after the vertical and horizontal innovation events in the survey. Moments about unconditional growth rates after vertical and horizontal innovation (relative to the average establishment) enter the calibration exercise as targeted moments in the calibration stage. These moments

[12]This concept was used in Akcigit and Kerr (2018) to measure the innovation within incumbents' own technology clusters.

inform how accurately vertical and horizontal survey answers are associated with the actual productivity increase and product line expansion. This unconditional growth rate may contain employment growth effects from other types of contemporaneous innovation. Additionally, innovation behavior before and after the innovation events are positively correlated with each other. Nonetheless, these moments are calculated in the same way across the model and the data.

The establishment-level employment growth rate over n periods before (-) and after (+) the innovation year is defined in a similar way as Davis et al. (1998).

$$EmpGr_{it}^{n-} = 2 \times \frac{AdmSize_{it} - AdmSize_{it-n}}{AdmSize_{it} + AdmSize_{it-n}}$$
$$EmpGr_{it}^{n+} = 2 \times \frac{AdmSize_{it+n} - AdmSize_{it}}{AdmSize_{it+n} + AdmSize_{it}},$$

where $AdmSize_{it}$ is the employment size of establishment i in year t reported via the administrative data set. Growth rates range from -2 to 2. Because of sample attrition issues, I only include establishments continuing in the sample. Thus, all entrant and exiting establishments, as well as missing samples, are excluded from the analysis.

To control for any time trend in employment at the establishment level, I normalize the growth rates of innovating establishments with the growth rates of all surviving establishments. I first calculate the means for $EmpGr_{it}^{n+,-}$ among all establishments in the sample. Then, I calculate the means for $EmpGr_{it}^{n+,-}$ among vertically or horizontally innovating establishments in year t. When taking means, each observation is weighted by the initial weight multiplied by their initial size[13]. I adjust the growth rates of innovating establishments by subtracting the average growth rate of surviving establishments in the sample. I further normalize the rates by subtracting the relative growth rates in the previous year of the innovation from all relative growth rates. Then, the (normalized) relative growth rate

[13]Initial weight or size means their weight or size in $t - n$ for $n-$, and their weight or size in t for $n+$.

one year before vertical or horizontal innovation becomes zero.

Figure 3: Cumulative Employment Growth before and after Innovation

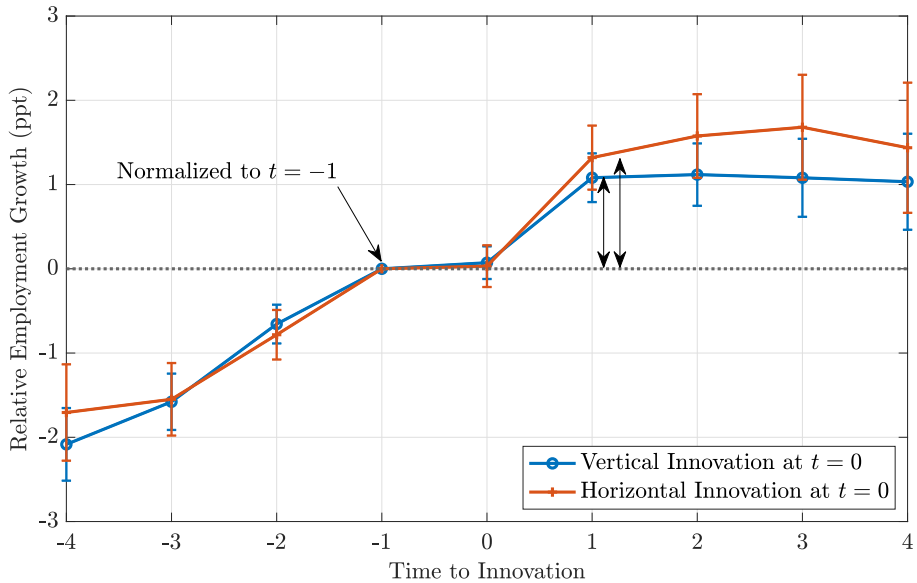


Figure 3 shows the unconditional relative employment growth rates over $n = -4, \ldots, 4$ before and after innovation events. The establishments already had steeper size growth than average establishments before the innovation events, suggesting establishments that choose to innovate are selective in terms of their growth trend. Appendix A.1 shows that establishments that innovate in year 0 are more likely to have additional innovation experiences in other years, implying the pre-trend in employment growth may come from previous innovation activities. However, the difference between growth rates of innovating establishments and average establishments becomes more distinct after the innovation year. Moreover, regression analyses in A.3 reveal that innovation events are still associated with establishment growth even after controlling for establishment fixed effects.

I use the growth rates between the previous year of the innovation and two years after the innovation. I do this to address the misalignment between the timing of innovation in the survey and the timing of employment measures in the administrative data set. Specifically,

the panel survey asks about innovation activities in the previous year of the survey, while employment measures are given as of June 30 of a given year. Therefore, it is ambiguous whether the innovation that implemented quality improvement in the year 2015 has employment measured on June 30, 2015, before or after the innovation event in 2015. Thus, while I call 2015 the innovation year, the growth rate in innovation year can measure employment growth before or after the innovation event. In the empirical exercise, I find that most size increases after innovation happen between the innovation year and the following year while there is no size growth between the innovation year and the previous year.

In Appendix A.3, I rigorously show the dynamic correlation of vertical and horizontal innovation events by introducing an event-study-like specification that assumes additive separability across the effects of innovation events over different types over years. The statistical model shows that vertical innovation has smaller and transitory effects on establishment-level employment while horizontal innovation has larger and more persistent effects. This is consistent with an assumption of the model that vertical innovation increases relative quality on existing product lines within establishments-its impact on employment is cumulative and depreciates rapidly as other establishments innovate and lower relative quality over time. On the other hand, horizontal innovation gives a new product line to the establishments, possibly after taking others' product lines. As a result, the employment increment after horizontal innovation is larger, and the size increment lasts longer because a loss of the additional product line happens rarely.

Additionally, Appendix A.4 shows what happens to non-innovating establishments when competitors in the same sector innovate more frequently. The regression analysis suggests that horizontal innovation is associated with lower employment growth rates and higher wage growth rates among non-innovators. This result is consistent with the interpretation that horizontal innovation leads to direct creative destruction and job reallocation in the economy.

2.3 Hartz Reform and Aggregate Moments

This chapter explains the details of the Hartz reform and shows aggregate time-series before and after the reform preceding the counterfactual analysis in Section 5. In the counterfactual exercise, I evaluate the impact of the reform on innovation and aggregate growth through the lens of the quantitative model constructed in Section 3. The Hartz reform is a large-scale labor market reform that started in the early 2000s. The reform took four stages (Hartz I, II, III, and IV). The first stage of the reform took effect on January 1, 2003, and the last stage of the reform came into effect on January 1, 2005.

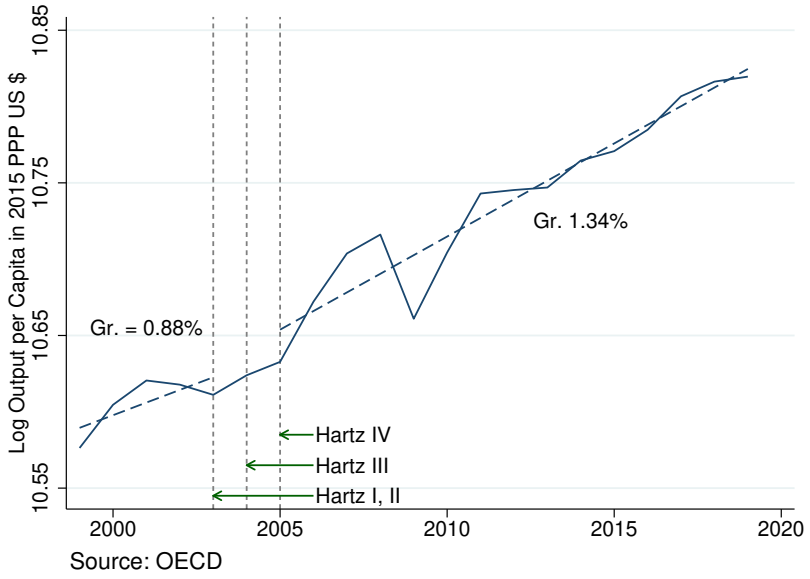
The Hartz reform had three cornerstones (Jacobi and Kluve, 2006). First, the reform improved the efficiency of government employment services. During the reform, government employment agencies were restructured, and the market-based mechanism was introduced to choose private employment agencies. Second, the reform incentivized unemployed workers to find a job by reorganizing the benefits system and requiring proactive actions from unemployed workers to receive the benefits. The reform also subsidized low-income wage earners. Lastly, the reform deregulated the labor market by expanding temporary work contracts and reducing firm size restrictions on dismissal protection.

The German economy indeed enjoyed a larger increase in output per capita and employment after the reform. As in Figure 4, after the Hartz reform I, II, III, and IV in the mid-2000s, the German economy grew faster than in the early 2000s. At the same time, the economy experienced a sharp increase in employment among workers aged 20-64 after the reform (see Figure 5). The increases in employment rates were even steeper because the population steadily decreased during that time. However, unlike the Schumpeterian predictions that more growth comes from establishment-level replacements[14], the job destruction rate decreased in Figure 6 during the periods of high aggregate output growth after the

[14]For more discussions, see Caballero (2010).

reform[15]. Consistent with the aggregate employment growth, more jobs were created after the reform. Even after the employment peaked and growth slowed down after 2014, the growth rates remained high, and the job destruction rates stayed low. This implies that the type of innovation that led to the fast economic growth in late 2014 was made by incumbent firms on their previous product portfolios, which did not require large job destruction at the establishment level.

Figure 4: GDP growth before and after the Hartz reform



[15]In the extensive margin, establishment exit rates also decreased during the same period.

Figure 5: Employment before and after the Hartz reform

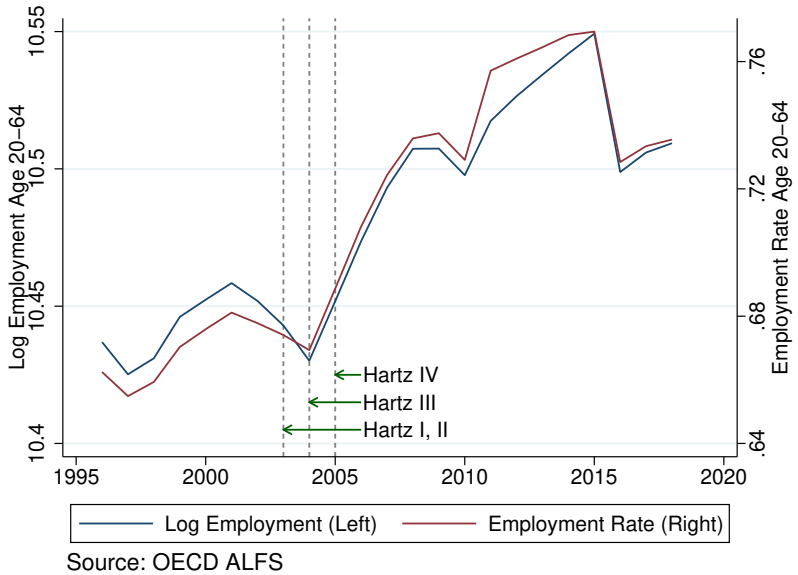
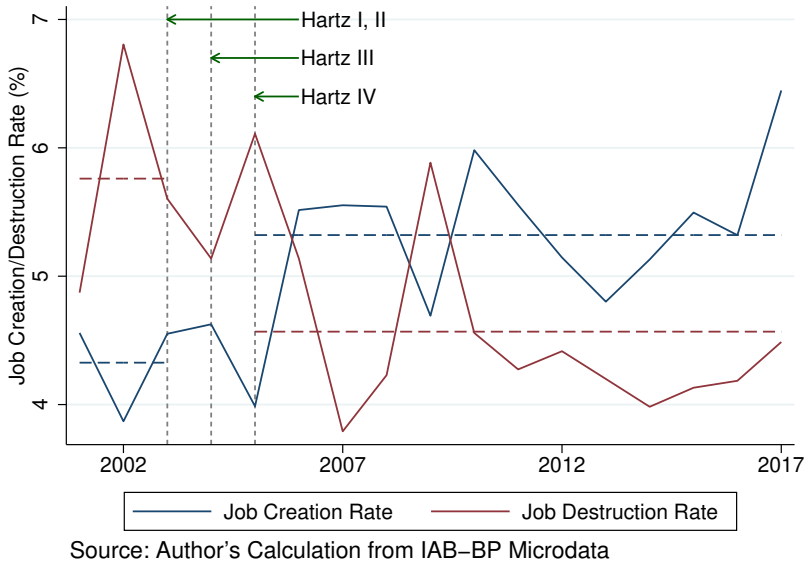


Figure 6: Job Creation and Destruction Rates



The following findings summarize key points that are incorporated in the calibration

and the counterfactual exercises.

1. Vertical innovation is more common than horizontal innovation.
2. Larger establishments are more likely to innovate both vertically and horizontally.
3. Employment grows after both vertical and horizontal innovation, but the effect is only slightly larger for horizontal innovation.
4. (Appendix A.4) Horizontal innovation is more strongly associated with relative job destruction of non-innovating establishments.
5. Labor market reform in Germany coincides with an increase in growth rates of output per capita as well as an increase in national employment. However, average job destruction rates decline after the reform.

3 Growth Model with Quality Ladders

The goal of the modeling exercise is to construct a framework to quantify the effects of vertical and horizontal innovation decisions at the firm[16] level on aggregate outcomes such as productivity growth and job turnover rates. Ultimately, the model is calibrated and used to quantify the macroeconomic effects of a major labor market reform in Germany. Vertical and horizontal innovation choices at the firm level are endogenized and aggregated to generate productivity growth as in Akcigit and Kerr (2018). The idiosyncratic innovation process at the product-line level generates endogenous dispersion of product qualities across production lines as in Acemoglu et al. (2018).

[16]The unit of observation in the empirical analysis is an establishment, but I call an establishment a 'firm' in the model.

3.1 Environments

The economy consists of representative households and both final and intermediate good producers. Representative households make a labor supply decision, gather all labor income and business profits, and decide a consumption path. Representative households consume homogeneous final goods, and the production of the final goods does not require any labor input. Final goods are competitively produced by combining differentiated intermediate goods. Each intermediate good variety corresponds to a product line and is produced by firms operating on the product line. Products on the same product line are perfectly substituted after adjusting their quality differences while products over different product lines are imperfectly substituted. Production of intermediate goods relies on labor inputs traded in the competitive and frictionless labor market.

Intermediate producers are heterogeneous with respect to their product line portfolios $\{x_j\}_{j=1}^{N_i}$, where N_i is the number of product lines held by the firm and x_j is the relative quality[17] of the product line. The revenue of this firm per unit of time is $\sum_{j \in N_i} \pi(x_j)$. Because of the aggregate growth, the relative productivity of each product line depreciates over time without any innovation behavior.

Firms can innovate vertically. Vertical innovation is specific to each product line. They choose a Poisson rate for vertical innovation, ξ, and hire $c(\xi)$ units of labor per unit of time on a product line. This labor cost is convex in the rate of vertical innovation, i.e. $c'_\xi(\xi), c''_\xi(\xi) > 0$. If they succeed in making such innovation on a product line, the relative quality of the product line jumps by $\lambda^i \bar{x}$ instantaneously, where the $\bar{x}$ is the cross-sectional mean of x. Vertical innovation does not affect the number of product lines held by the firm, N_i.

[17] This can also be interpreted as the relative productivity because of the isomorphism between quality and inverse productivity in the Constant Elasticity of Substitution (CES) demand structure with the Constant Return to Scale (CRS) technology.

Firms also innovate horizontally with a Poisson rate ζ at a labor cost of $wc_\zeta(\zeta)$ per unit of time on a product line. The cost of innovation is also convex, i.e. $c'_\zeta(\zeta), c''_\zeta(\zeta) > 0$. Conditional on the horizontal innovation event, a firm is matched to a vacant product line with probability p. The productivity of the firm on the new product line is simply the current productivity of the product line the horizontal innovation is based on. Without any competitor, the firm starts producing on the vacant line.

With probability $1-p$ conditional on horizontal innovation, firms are matched with a product line that is currently occupied by another firm. I assume that only the best quality producer operates on the product line at the monopolistic price, and I rule out the probability of limit-pricing where the best quality producers charge the price that makes the second-best quality producer break even. This behavior can be micro-founded by the two-stage Bertrand game as in Akcigit and Kerr (2018)[18]. Then the expanding firm can successfully catch the production opportunity only if it is more productive than the current occupier of the matched product line. Let $F(x)$ denote the probability that the firm expanding on a product line with productivity x is more productive than randomly matched product lines. With undirected random matching, $F(x)$ is the cumulative distribution function of the relative product line quality, x. As a result, $p + (1-p)F(x)$ is the probability that the firm gains a new product line after horizontal innovation.

The firm loses a product line when either an expanding firm or an entering firm is matched to the product line and is more productive. The Poisson rate of this event is denoted by $\tau(x)$, and $\tau'(x) > 0$. For simplicity, this loss of a product line is assumed to be permanent. Thus, even if the producer with the best quality on the line disappears, the producers with lower qualities cannot operate on the product line again.

[18] In the two-stage Bertrand game, all producers on the product line pay $\epsilon > 0$ to proceed to the second stage in the first stage. In the second stage, only the firms that paid in the first stage can set their price and sell in the market. In the Subgame Perfect Nash Equilibrium, only the best quality producer pays in the first stage and sets monopolist price in the second stage.

A firm exits the market in the two cases. First, a firm exits the market at an exogenous rate δ. In this case, all product lines held by the exiting firm are left vacant and subject to being matched with expanding and entering firms instantaneously. The probability of being matched with a vacant product line after horizontal innovation, p, is determined by the mass of exogenously exiting firms versus the mass of expanding incumbents and entrants. Second, a firm exits the market if the firm had only one product line and if the last product line is poached by another firm.

Entering firms start with one product line. An entering firm first randomly meets a product line to learn and makes vertical innovation with a step size $\lambda^e \bar{x}$. It also meets a vacant product line with the same probability as the horizontal innovation case, p, and starts producing on the matched product line. With probability $1-p$, it meets an occupied product line. It can produce on the occupied product line if and only if it is more productive than the firm currently operating on the product line.

Later, I focus on a balanced growth path (BGP) equilibrium where the interest rate is constant, the final output and the wage grow deterministically at a constant rate g_Y, the relative productivity of a product line depreciates at a constant rate g_x, and the value and the distribution over x are time-invariant. Thus, the time subscript t is omitted when there is no confusion.

3.2 Values

Again, the state variable of a firm is the set of productivity levels on product lines it occupies. Since all decisions are product line specific and the revenues from product lines are additively separable, the total firm value is the sum of product line values. Furthermore, on the BGP, the value of a product line is a product between the time-invariant value that only depends on the relative quality and the aggregate output. Appendix B.1 shows the formulation of firm-level value function as well as the multiplicative separability of product-line

values between the aggregate component Y_t and the time-invariant idiosyncratic component $v(x_j)$ that only depends on the relative quality. Thus, the firm value $\mathfrak{V}_t\left(\left\{x_j\right\}_{j=1}^{N_i}\right)$ can be expressed as in the following equation.

$$\mathfrak{V}_t\left(\left\{x_j\right\}_{j=1}^{N_i}\right) = Y_t \sum_{j=1}^{N_i} v\left(x_j\right),$$

where $v(x)$ is the time-invariant component of the product line value and given by the following Hamiltonian-Jacobi-Bellman equation.

$$\begin{aligned}\left(\rho+\delta+\tau\left(x\right)\right) v\left(x\right) = \max_{\xi,\zeta\geq 0}\Bigg\{ & \underbrace{\pi\left(x\right)}_{\text{profit}} - \underbrace{g_x x v'\left(x\right)}_{\text{depreciation}} \\ & - wc_\xi\left(\xi\right) + \xi \underbrace{\left(v\left(x+\lambda^i\bar{x}\right) - v\left(x\right)\right)}_{\text{return to V.I.: better quality}} \\ & - wc_\zeta\left(\zeta\right) + \zeta \underbrace{\left(p+(1-p)F(x)\right) v\left(x\right)}_{\text{return to H.I.: additional line}} \\ & - \left(\tau\left(x\right)+\delta\right)\kappa x\Bigg\}.\end{aligned}$$

In this equation, ρ is the discount rate[19], $w = W_t/Y_t$ is the wage rate relative to the final output (which is constant at over the BGP), δ the exogenous exit rate, $\tau(x)$ the creative destruction rate, $\pi(x)$ the revenue defined below, $c_\xi(\xi)$ the vertical innovation cost, and $c_\zeta(\zeta)$ the horizontal innovation cost.

Vertical innovation increases the relative quality of the product line with a constant

[19]This is the same as the discount rate of the representative household, which is described below.

step size, $\lambda^i \bar{x}$. Horizontal innovation, adds a new product line with the same productivity by probability $p + (1-p)F(x)$. The horizontal innovation rate is unambiguously increasing in x because the value is increasing in x. Vertical innovation is not monotone in x but tends to increase in x when x is low because the potential return to horizontal innovation is increasing faster with a higher probability of a successful expansion through a higher $F(x)$.

$\kappa \geq 0$ represents the cost of unexpected product line closures the firm must pay to fire workers. This is empirically related to severance pay packages and administrative costs the firm must pay to end the contract earlier than the agreed date. I assume $\kappa = 0$ in the baseline case but $\kappa > 0$ in one of the counterfactual scenarios. Section 5.1 contains more details about the interpretation of the reform in the context of the model.

3.3 Entry

Like vertical and horizontal innovation, entrant innovation also requires labor inputs. Entrants must hire $c(\chi)$ units of labor to enter the market and draw initial productivity as a fixed cost of entry. The free entry condition determines the rate of firm entry, χ.

$$wc_\chi(\chi) = \chi\left(\int (p + (1-p)F(x + \lambda^e \bar{x}))v(x + \lambda^e \bar{x})dF(x)\right).$$

In the free entry equation, $c_\chi(\chi)/\chi$ is a cost per entry and increasing in χ. Entrant innovation also requires labor inputs, and the average cost of entrant innovation is increasing in equilibrium entry rate.

3.4 Innovation and Job Turnover

The firm-level employment is expressed as the sum of employment size over product lines held by the firm, as in the following equation.

$$\sum_{j=1}^{N_i} l(x_{jt}) := \sum_{j=1}^{N_i} \{ \underbrace{\frac{\sigma-1}{\sigma}\frac{x_{jt}}{w_t}}_{\text{prod. labor}} + \underbrace{c_\zeta(\zeta(x_{jt})) + c_\xi(\xi(x_{jt}))}_{\text{R\&D labor}} \}$$

The employment on each product line, $l(x)$, has two components: production and R&D labor. The production labor is linearly increasing in the relative quality x_{jt}. The R&D labor for horizontal innovation $c_\zeta(\zeta(x_{jt}))$ is also increasing in x_{jt} because the return to horizontal innovation is increasing in relative quality. The vertical innovation rate does not have an unambiguous relationship to x_{jt} but numerically tends to increase in x_{jt} because of higher returns to horizontal innovation.

If vertical innovation happens to a product line with x, the innovating firm on the product line creates the following amount of new jobs.

$$l(x+\lambda^i\bar{x}) - l(x) = \lambda^i\bar{x}\frac{(\sigma-1)}{w\sigma} + (c_\zeta(x+\lambda^i\bar{x}) - c_\zeta(x)) + (c_\xi(x+\lambda^i\bar{x}) - c_\xi(x))$$

First, the demand curve for the firm with better quality shifts to the right, so the firm increases production labor by$\lambda^i\bar{x}(\sigma-1)/\sigma$. Then, the returns to horizontal and vertical innovations change. Consequently, the R&D labor input also changes. As mentioned above, horizontal R&D inputs unambiguously increase, but vertical R&D inputs may either increase or decrease. Still, because of production and horizontal R&D labor demand increasing in x, more jobs are made after vertical innovation. Non-innovating establishments in the economy reduce their labor input because of the depreciation of their relative quality.

Suppose that a product line experiences horizontal innovation with the same productivity increase from x to $x+\lambda^i\bar{x}$. Then, the new firm on the product line creates $l(x+\lambda\bar{x})$

amount of new jobs, while the old firm loses $l(x)$ amount of labor. Other non-innovating establishments experience the same degree of relative quality depreciation because the change in average quality remains the same. Thus, horizontal innovation generates an additional $l(x)$ amount of job turnovers across firms.

On the BGP, for a sufficiently small time interval Δ, establishments create jobs in three cases[20]: vertical innovation, horizontal innovation, and establishment entries. Vertically innovating establishments increase employment by $l(x + \lambda^i \bar{x}) - l(x) - \Delta g_x x l'(x)$, horizontally innovating establishments increase employment by $l(x) - \Delta g_x x l'(x)$, and entrant establishments create jobs by $l(x + \lambda^e \bar{x})$. The quality depreciation term, $\Delta g_x x l'(x)$, applies to all incumbents regardless of their innovation status. Then, the job creation at the product line level can be approximated as the following equation.

$$\begin{aligned} JC := & \int \Delta\xi(x)(l(x + \lambda^i \bar{x}) - l(x) - \Delta g_x x l'(x))dF(x) \\ & + \int \Delta\zeta(x)(p + (1-p)F(x))(l(x) - \Delta g_x x l'(x))dF(x) \\ & + \int \Delta\chi(p + (1-p)F(x + \lambda^e \bar{x}))l(x + \lambda^e \bar{x})dF(x) \end{aligned}$$

Thus, increases in $\xi(x)$ and $\zeta(x)$ increase more jobs directly in the equation. However, more innovations also increase g_x, which accelerates the depreciation of relative quality and reduces job creation by innovating firms. Quantitatively, for the same increase in g_x, horizontal innovation has larger effects on job creation than on vertical innovation. Thus, horizontal innovation is more likely to increase the aggregate amount of job creation.

Establishments experience job destruction in three cases: exogenous exits, endogenous creative destruction, and no establishment-level shock. An exogenous exit and a creative destruction shock destroy jobs by $l(x)$. Establishments without any innovation or shock

[20] For a small Δ, only one of the events happens to a product line. In this case, I set $\Delta = 1$.

also experience job destruction by $\Delta g_x x l'(x)$ because of endogenous quality depreciation. Then, the job destruction at the product line level can be approximated as the following equation.

$$JD := \int \Delta(\tau(x) + \delta) l(x) dF(x) + \int (1 - \Delta\bar{\zeta} - \Delta\bar{\xi} - \Delta\bar{\chi} - \Delta\bar{\tau} - \Delta\delta) \Delta g_x x l'(x) dF(x)$$

In this equation, $\Delta\bar{\zeta} := \int \Delta\zeta(x)(p + (1-p)F(x))dF(x)$ is the share of horizontally expanding product lines. Likewise, $\Delta\bar{\xi} := \int \Delta\xi(x)dF(x)$ is the share of vertically innovating product lines, $\Delta\bar{\chi} := \Delta\chi \times \int p + (1-p)F(x + \lambda^e \bar{x}))dF(x)$ is the share of product lines improved by entrant establishments, $\Delta\bar{\tau} = \int \Delta\tau(x)dF(x)$ is the share of product lines hit by creative destruction shocks, and $\Delta\delta$ is the share of product lines that belong to exiting establishments.

On the BGP, the aggregate labor supply is constant, which implies $JC \approx JD$. In the transition path where labor supply changes, JC does not have to be equal to JD. Finally, the job creation and destruction rates are defined as JC and JD divided by $L_t - c_\chi(\chi)$ (the aggregate labor hired by incumbents[21]), respectively. On the BGP, job creation and destruction rates are equal. Thus, I call the single rate the job reallocation rate in Table 4.

3.5 Final Good Production

Homogeneous final goods are competitively produced by aggregating intermediate goods over different product lines with constant elasticity of substitution. The final good is the

[21] It is ambiguous how the R&D labor hired for entrant innovation is mapped into the employment variable in the data. In the end, I exclude them from the model-generated observations.

numeraire.

$$Y_t = \left(\int_0^1 a_{jt} y_{jt}^{\frac{\sigma-1}{\sigma}} dj \right)^{\frac{\sigma}{\sigma-1}},$$

where $\sigma > 1$ is the elasticity of substitution, a_{jt} is the quality of the intermediate good from product line j at time t, and $j \in [0, 1]$ is the normalized set of product lines. As described before, intermediate inputs on the same product line are perfectly substituted up to quality units while intermediate inputs over different product lines are imperfectly substituted with a finite elasticity of substitution, σ. As a result, only the firm with the highest quality operates on a product line in equilibrium.

After the two-stage Bertrand game described above, the firm sets a price that maximizes the revenue taking the demand of final good producers as given. The quality of the firm shifts the demand curve given the same price but does not affect the demand elasticity thus has no impact on the mark-up. The profit maximization problem is described in the following equation.

$$p_{jt} = \arg\max_{p_{jt}} \left\{ y_{jt} p_{jt} - W_t y_{jt} \right\},$$

subject to

$$y_{jt} = a_{jt}^{\sigma} Y_t P_t^{\sigma} \left(p_{jt} \right)^{-\sigma}$$

where $P_t := \left(\int_0^1 a_{jt}^{\sigma} p_{jt}^{1-\sigma} dj \right)^{\frac{1}{1-\sigma}}$ is the quality-adjusted average input price index and W_t is the wage for unit labor. In equilibrium, $P_t = 1$ for zero-profit of final good producers. The labor productivity is normalized to one on all product lines so that one labor unit is converted to one unit of intermediate goods[22]. The monopolists' pricing follows a simple

[22]This specification is isomorphic to a case with production efficiency improvements. For this reason, I

constant mark-up rule, $p_{jt} = \sigma/(\sigma-1)W_t$. In equilibrium where all prices are set by this rule, the zero-profit condition for final good producers can be written as

$$\frac{\sigma}{\sigma-1}W_t = A_t := \left(\int_0^1 a_{jt}^{\sigma} dj\right)^{\frac{1}{\sigma-1}}.$$

Thus, in the equilibrium, the wage must increase with the average quality index in the economy. Moreover, the labor market demand aggregation yields the following equation.

$$L_t^p := \int l_{jt} dj = Y_t A_t^{-1}.$$

L_t^p is the aggregate labor demand for production. Thus, for a fixed production labor input, Y_t increases linearly in A_t. Consequently, in a BGP equilibrium where the quantity of labor supply is fixed and Y_t increases at a constant rate, W_t increases linearly in Y_t. In this case, I can define the value-added only as a function of relative quality, x_{jt}.

$$\pi_t(x_{jt}) = Y_t \pi(x_{jt}) := Y_t \left(\frac{1}{\sigma}\right) x_{jt}.$$

In this function, $x_{jt} := a_{jt}^{\sigma}/A_t^{1-\sigma}$.[23] It can be further shown that the production labor input demand is also linear in x_{jt}.

Vertical innovation improves the quality of the product line with a_{jt} by some step size so that the innovation step size is constant in relative quality x_{jt}. See Appendix B.2 for the explanation and the mapping between the step sizes. Appendix A.3 further provides rationales for the constant step size for vertical innovation and constant productivity on the new product line after horizontal innovation.

included both quality improvement and process innovation into my definition of vertical innovation in the empirical analysis.

[23] This definition of x_{jt} implies that $\bar{x} := \int x_{jt} dj = 1$. Moreover, this definition implies a constant depreciation rate of x_{jt} at rate $g_Y \times (\sigma - 1)$ without any changes in a_{jt} on the BGP.

3.6 Representative Household

The household block follows a standard-setting in the Schumpeterian growth model literature such as Acemoglu et al. (2018). The representative household gathers all labor incomes and profits of firms together and determines a consumption path, given the sequences of wage $\{W_t\}_{t\geq 0}$ and real interest rates $\{r_t\}_{t\geq 0}$. The representative household maximizes the following lifetime utility:

$$\max_{C_t} \int_0^\infty e^{-\rho t} \left(\log(C_t) - \frac{\phi L_t^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} \right) dt$$

subject to

$$C_t + \dot{b}_t = W_t(1-\omega_L)L_t + \int_0^1 \pi_{jt} dj + r_t b_t + \omega_L W_t \tilde{L}_t$$

where the right-hand side is the total income of the household and b_t is the real bond holding with zero net supply. The income of the household is the sum of all the labor earnings ($(1-\omega_L)W_t L_t$), the profits net of innovation costs (π_{jt}), the interest rate of real bond holdings, and the lump-sum transfer from labor income tax $\omega_L W_t \tilde{L}_t$, where ω_L is the labor income tax rate or the labor wedge that affects the labor supply decision of the representative household. In equilibrium, $L_t = \tilde{L}_t$ but the representative household does not take into account the effect of its own labor supply decision on the lump-sum transfer. The ω_L is assumed to be zero in the baseline case but set positive in one of the counterfactual scenarios below. In equilibrium, $C_t = Y_t$.

The Euler equation of the lifetime utility maximization is as follows:

$$\frac{\dot{C}_t}{C_t} = r_t - \rho.$$

On the balanced growth path where $C_t = Y_t$ grows at g_Y, the interest rate is simply

$$r_t = r = \rho + g_Y.$$

Next, the first order condition with respect to labor supply gives a labor supply curve.

$$\phi \left(L_t^s\right)^{\frac{1}{\eta}} = \frac{W_t(1-\tau_L)}{C_t} = w\left(1-\tau_L\right).$$

Again, $w := W_t/Y_t$ is constant in the BGP equilibrium.

3.7 Distribution

The distribution of product lines across x is characterized by the following equation.

$$\begin{aligned} g_x x f(x) &= \delta F(x) + \int_{s\in(x-\lambda^i\bar{x},x]} \xi(s)dF(s) \\ &\quad + (1-p)(G(\infty)-G(x))F(x) \\ &\quad - pG(x) \\ G(x) &:= \int_{s\leq x} \zeta(s)dF(s) + \int_{s\leq x-\lambda^e\bar{x}} \chi dF(s). \end{aligned}$$

The total measure of product lines is normalized to one. For derivation, see Appendix B.3.

The probability of expanding and entering firms to meet a vacant line, p, is given by the condition that the mass of product lines left vacant because of exogenous exit is equal to the mass of expanding and entrant firms that are matched to vacant lines.

$$\delta = p\int \zeta(s)dF(s) + \chi p$$

The creative destruction rate $\tau(x)$ at the firm level must satisfy the following equation that implies the rate at which incumbent firms are taken out from their product lines by creative destruction is equal to the rate at which expanding firms poach new product lines.

$$\tau(x) = (1-p)\left(\chi\left(\int_{s>x-\lambda^e\bar{x}} dF(s)\right) + \int_{s>x} \zeta(s)dF(s)\right)$$

The depreciation rate for x is given by the following equation. The equation decomposes the aggregate growth rate into three margins related to vertical innovation, horizontal innovation, and entrant innovation. g_Y is simply $g_x/(\sigma-1)$.

$$\begin{aligned}
g_x\bar{x} = & \underbrace{\int\left[\int_{s\in(x-\lambda^i\bar{x},x]} M\xi(s)dF(s)\right]dx}_{\text{vertical innovation}} \\
& + p\underbrace{\int\left[\int_{s>x} M\zeta(s)dF(s)\right]dx - p\bar{\zeta}\bar{x}}_{\text{horizontal innovation (1)}} \\
& + p\underbrace{\int\left[\chi\int_{s>x-\lambda^e\bar{x}} dF(s)\right]dx - p\chi\bar{x}}_{\text{entrant innovation (1)}} \\
& + (1-p)\underbrace{\int\left(\int_{s>x} M\zeta(s)dF(s)\right)F(x)dx}_{\text{horizontal innovation (2)}} \\
& + (1-p)\underbrace{\int\left(\chi\int_{s>x-\lambda^e\bar{x}} dF(s)\right)F(x)dx}_{\text{entrant innovation (2)}}
\end{aligned}$$

3.8 Labor Market Clearing

The final good market is cleared if $C_t = Y_t = A_tL^p$, where L^p is the production labor inputs. Labor inputs are also used to implement vertical, horizontal, and entrant innovation.

The total labor demand is given by the following equation.

$$W_t L_t = \frac{\sigma - 1}{\sigma} Y_t + W_t \int_0^1 c_\xi(\xi_{jt}) dj + W_t \int_0^1 c_\zeta(\zeta_{jt}) dj + W_t c_\chi(\chi_t)$$

On the BGP, the wage and the final good grow at the same rate. Thus, the labor market clearing condition becomes

$$wL = \frac{\sigma - 1}{\sigma} + w \int_0^1 c_\xi(\xi_{jt}) dj + w \int_0^1 c_\zeta(\zeta_{jt}) dj + w c_\chi(\chi_t).$$

3.9 Equilibrium

The Balanced Growth Path (BGP) equilibrium is defined as value function $v(x)$; policy function $\xi(x)$, $\zeta(x)$; product line distribution $f(x)$; creative destruction rates $\tau(x)$; the probability of meeting a vacant product line p; depreciation and growth rates g_x, g_Y; the entry rate χ; and a constant wage rate relative to the output w; a constant interest rate r; and the constant labor supply L such that

1. The representative household optimizes[24].
2. Value function and policy function satisfy the HJB equations.
3. The distribution of product lines over x stays constant. The mass of product lines is also time-invariant[25].
4. The free entry condition is met to pin down the entry rate.
5. Growth and creative destruction rates are internally consistent.

[24] The representative household's optimization fixes the interest rate over the BGP and determines the labor supply curve.

[25] If $\zeta(x)$ and χ are too small, it is possible that the mass of available product lines may get lower over time because of net product line loss after exogenous firm exits. Here I am restricting the value of c_χ and c_ζ such that this does not happen and the mass of available product lines is constant over time.

6. The probability of meeting a vacant line is such that the mass of product line is constant over time.

7. Final goods, intermediate goods, and labor markets are cleared.

4 Calibration

For numerical illustration, the unit of the time is set to months. The innovation and entry cost functions are parameterized as iso-elastic functions of innovation and entry rates, respectively.

$$c_\xi(\xi) = c_\xi \frac{\epsilon_\xi}{\epsilon_\xi + 1} \xi^{\frac{\epsilon_\xi + 1}{\epsilon_\xi}}$$
$$c_\zeta(\zeta) = c_\zeta \frac{\epsilon_\zeta}{\epsilon_\zeta + 1} \zeta^{\frac{\epsilon_\zeta + 1}{\epsilon_\zeta}}$$
$$c_\chi(\chi) = c_\chi \chi^{\frac{\epsilon_\chi + 1}{\epsilon_\chi}}$$

To match the small size growth after vertical and horizontal innovation, two additional parameters are added in the calibrated version of the model. The parameters, $q^v, q^h \in [0, 1]$, govern the probabilities that productivity increases and the expansion to a new product line are successful conditional on positive answers to the survey questions on vertical and horizontal innovation. Then, the returns to vertical and horizontal innovation become simply $q^v \xi \left(v\left(x + \lambda^i \bar{x}\right) - v\left(x\right)\right)$ and $q^h \zeta \left(p + (1-p)F(x)\right) v\left(x\right)$.

The parameters are calibrated through the following weighted minimization procedure:

$$\hat{\Omega} = \arg\min (\bar{m} - m(\Omega))' W (\bar{m} - m(\Omega)),$$

where $\Omega = (\lambda^i, \lambda^e, \sigma, c_\xi, \epsilon_\xi, c_\zeta, \epsilon_\zeta, \chi, q^v, q^h)$ is a vector of internally calibrated parameters, $\bar{m}$

is a vector of empirical moments in Table 2, $m(\Omega)$ is the counterpart moments from the simulated model given Ω, and W is a weighting matrix.

The moments in Table 1 are transferred directly into the numerical model. The discount rate of the representative household, ρ, is calibrated such that the interest rate $r = \rho + g_Y$ is equal to 8% a year. The exogenous exit rate, δ, comes from Fackler et al. (2013) to match the exit rate of establishment with an employment size of 100 or more. The quantity of labor supply in the steady state is calibrated to normalize the wage level in the steady state to one.

Table 1: Externally Calibrated Moments

Moment	Meaning	Value	Source
r	Discount Rate (%)	8/12	Average Stock Market Rate of Return
δ	Exogenous Exit Rate (%)	1/12	Fackler et al. (2013)
$\bar{L}$	Labor Supply in the Steady State	0.38	Wage Normalization

The moments in Table 2 are matched after solving the equilibrium and simulating the panel survey in a way consistent with the German survey[26]. The model is exactly identified with 10 moments to match with 10 parameters. Overall the model can match the moments closely, despite the uncertainty associated with Monte Carlo simulations. Table 3 shows the values of calibrated moments.

[26] I start with 50,000 establishments each with a single product line, burn in 100 years, and simulate the panel until it converges in terms of i) the fraction of vertically innovating establishments, and ii) the fraction of horizontally innovating establishment. The convergence criterion is 1e-4.

Table 2: Internally Calibrated Moments

Moment	Data	Model	Target
Growth Rate (%)	1.3	1.3	λ^i
Fraction of Vertical Innovation (%)	29.4	29.4	c_ξ
Fraction of Horizontal Innovation (%)	17.3	17.4	c_ζ
Fraction of Vertical Innovation (Medium - Small, ppt)	3.6	3.4	ϵ_ξ
Fraction of Horizontal Innovation (Medium - Small, ppt)	2.0	1.3	ϵ_ζ
Employment Growth (Vertical - Average, ppt)	1.1	1.1	q^v
Employment Growth (Horizontal - Average, ppt)	1.3	1.4	q^h
Employment Growth (No Innovation - Average, ppt)	-1.6	-1.4	σ
Employment Share of Establishment Age ≤ 1 (%)	1.1	1.5	λ^e
Establishment Share of Establishment Age ≤ 1 (%)	1.9	1.9	χ

Note: The Medium and the Small groups of establishments mean the first and the second tertile groups in the size distribution at the sector level after trimming establishments at the 10th size decile, respectively. All size tertile groups then are pooled across industries. All moments are constructed in the same way as the survey. Employment growth rates are measured over two years conditional on survival. Vertical-Average means the employment growth after vertically innovating establishments relative to the average survived establishments, and other relative rates are defined similarly. All observations are weighted by the cross-sectional weights from the data provider except the size growth rates. Size growth rates are weighted by the cross-sectional weight multiplied by the lagged size.

Table 3: Values of Calibrated Parameters

	Parameter	Value
λ^i	Step Size of Vertical Innovation (%)	2.7
λ^e	Step Size of Entrant Innovation (%)	3.1
σ	Elasticity of Substitution across Products	1.4
c_ξ	Vertical Innovation Cost Shifter	2039
ϵ_ξ	Vertical Innovation Cost Elasticity	0.53
c_ζ	Horizontal Innovation Cost Shifter	2.61e+06
ϵ_ζ	Horizontal Innovation Cost Elasticity	0.31
$12 \times \chi$	Annual Share of Entrant Draws (%)	0.92
q^h	Probability of Successful Expansion (%)	3.1
q^v	Probability of Successful Improvement (%)	55

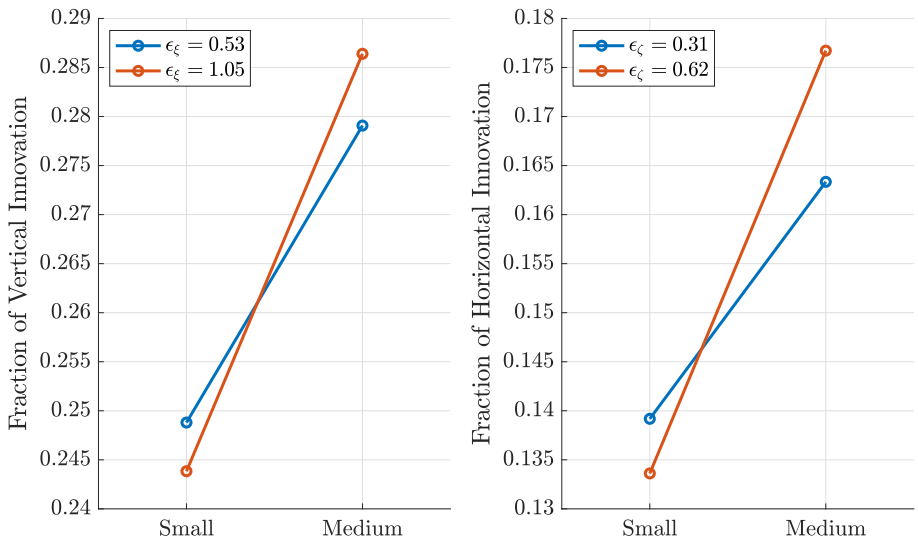
The step sizes of vertical and entrant innovation λ^i and λ^e jointly target the output growth rate on the BGP. λ^e further matches the employment share of entrants given the count share of entrant establishments. The idea is, as the step size of entrant innovation gets higher, entrants become more productive on average, thus their share in total employment is large at the given establishment share. The count share of establishment is met by the flow rate of entrant draws, χ. On the BGP, the elasticity of entry is not identified with the single flow rate. As a result, I first calibrate χ, back out c_χ from assumed values of ϵ_χ, and implement counterfactual analyses below. The elasticity of substitutions across product lines σ is calibrated to match the employment growth of non-innovating establishments relative to an average establishment. If σ is higher, non-innovating establishments shrink at a higher rate because of continuous substitution towards better-quality intermediate goods. The probability of successful quality improvement, q^v, is aimed to match the 2-year size growth rate of establishments after the vertical innovation relative to the average 2-year size growth rate of all continuing establishments. Likewise, the probability of successful expansion after horizontal innovation, q^h, is to match the 2-year size growth rate after horizontal innovation relative to vertical innovation.

The moments already predict a low value for q^h. Assume no exit and no entry for simplicity. Then, on average, continuing establishments' growth rate is close to zero. If an establishment with one product line experiences successful horizontal innovation, the establishment growth rate must be about 100%. If an establishment with two product lines experiences successful horizontal innovation on one of its product lines, the size growth rate must be 50%. The growth rate after horizontal innovation, relative to the average establishment, is only about 1.5%. This low number implies that most horizontal innovation in the survey fails to yield an additional product line.

The parameters of particular interest for the counterfactual analysis below are the elasticities of vertical and horizontal innovation, ϵ_ξ and ϵ_ζ. If the values of these elasticities are

high, the general equilibrium effects of some policy changes are more amplified by endogenous responses in incumbent innovation. On the other hand, with lower values of elasticities, the margin of endogenous incumbent innovation is quantitatively unimportant. Then, the cost-shifters of vertical and horizontal innovation, c_ξ and c_ζ match the average fraction of establishments experiencing innovation.

Figure 7: Identification of Innovation Elasticity



I indirectly infer the values of the innovation elasticities through the size profile of innovation frequencies. The two panels of Figure 7 show the frequencies of vertical and horizontal innovation across equally sized establishment bins. The 'small' bin corresponds to a group of establishments below the third decile of the size distribution, and the 'medium' bin corresponds to a group of establishments in the next three deciles of the size distribution. The largest establishments in the last four deciles are excluded from calculating moments to match because the model over-predicts the largest establishments' innovation frequencies after assuming independent innovation events across product lines. Larger establishments are more likely to innovate for two reasons. First, for establishments with the same number of product lines, establishment size is a predictor for the underlying productivity, and better

quality establishments are more likely to innovate. Moreover, larger establishments tend to own multiple product lines, and multi-product establishments are more likely to innovate on at least one of their product lines with mutual independence of innovation events across product lines. Ideally, small establishments are more likely to have a single product line, and the difference among small-sized establishments is more likely to capture the dependency on the underlying productivity and inform the model about how elastically those innovations respond to the underlying productivity differences.

In Figure 7, a higher value of innovation elasticity is associated with a steeper profile of innovation frequencies over employment size among small establishments between the first deciles and the second three deciles mostly due to the underlying difference in productivity[27]. For the same values of vertical and horizontal elasticity, horizontal innovation has a steeper profile across employment bins. This result comes from the assumption that the return to horizontal innovation is increasing more sharply in current productivity because productivity on the new product line is the same as the old product line that the horizontal innovation is based on. Since the model assumes the same step size of vertical innovation for all product lines[28], vertical innovation does not have a steep size profile. Empirically, however, size dependency is similar in proportion between vertical and horizontal innovation, implying that vertical innovation reacts more elastically to underlying productivity differences.

Appendix B.4 discusses the model fit by comparing moments untargeted in the calibration exercise. The calibrated model matches qualitative trends in employment growth before and after innovation events, fractions of innovating establishments across detailed lagged size groups, and job creation and destruction rates overall decreasing in lagged size of establishments.

[27]Notice that I further adjust the values of c_ξ and c_ζ to match the average fraction of innovating establishments.

[28]The validity of these assumptions is covered in Appendix A.3.

5 Application: the Hartz Reform and Innovation

I consider two counterfactual scenarios. The first scenario deals with the 'Hartz Reform', a major labor market reform that happened in Germany during the mid-2000s. I interpret the reform as a shift of the labor supply curve to the right and a decrease in the cost of unexpected product line closures. The second scenario addresses the new R&D tax credit policy that was introduced in Germany in 2020. I focus on the first scenario of Hartz reform and set the second scenario aside in Appendix C.3.

For each scenario, I first compare the new BGP under the environmental change to the baseline BGP. Aggregate productivity growth and job creation rates are decomposed into the three margins on vertical, horizontal, and entrant innovation under the two different BGP equilibria. Then, I evaluate changes in aggregate moments overt the transition path. I start from the baseline BGP, introduce an unexpected policy or environment change at $t = 0$, and compute the deterministic transition path to the new BGP with the change. The equations used to calculate the transition path equilibrium are expressed in Appendix C.1.

5.1 Interpretation of the Hartz Reform

I study the effect of the Hartz reform in Germany on the aggregate economy. The Hartz reform is a large-scale labor market reform that started in the early 2000s. For institutional components, see Section 2.3. In this section, I explain how the measures of the reform are translated in the context of the model around two dimensions.

To start with, I interpret one effect of Hartz reform as an exogenous increase in labor supply through a decrease in labor wedge in the representative household's problem. This interpretation is associated with the modernization of federal employment agencies and improving work incentives. The values for labor wedge before and after the Hartz reform are assumed to be strictly positive and zero, respectively. The value of the labor wedge in

the pre-reform period is set to match the increase in labor supply between the early 2000s and the late 2010s. According to the Annual Labour Statistics by the OECD, the average national employment of the population aged 20-64 between 2000-2004 was 34 million in Germany, and it grew to 37 million (7.6%) in the late 2010s. Meanwhile, that same German population (age 20-64) decreased by 2.5%, from 51 million to 50 million. I interpret the 10 percentage points additional increase in employment as the effect of Hartz reform and set the labor wedge before the Hartz reform to match this increment. The baseline Frisch elasticity of labor supply is assumed to be one, but I test other values for robustness in Appendix C.2. Under this parameter value, the implied labor wedge before the reform is 17%.

Second, I interpret the other effect of reform as lowering the cost of unexpected product line closures, which is associated with severance pays and administrative firing costs to the workers. For simplicity, I assume that the closure cost increases linearly in x and does not depend on the actual employment[29]. This cost is summarized as κ in the model. The severance pay is around 50% of the monthly wage per tenure year, and the mean tenure of a German worker is 10 years. There are not many studies about estimating firing costs in Germany, but a paper by Bentolila and Bertola (1990) estimated that the firing cost corresponds to 75% of annual wages in Germany. A product line with x hires $\frac{\sigma-1}{\sigma}x/w$ units of production labor and $c_\zeta(\zeta(x)) + c_\xi(\xi(x))$ amount of R&D labor. Assuming R&D labor demand is proportional to x as well, $\kappa = (10 \times 0.5 + 0.75 \times 12) \times L \approx 4.99$ before the reform, and $\kappa = 0$ after the reform.

Like the Frisch labor supply elasticity, entry elasticity c_χ is not identified in the BGP with the single free entry condition. However, the employment share of entrants stagnated

[29] If the cost of unexpected product line closures depends on the actual employment, the establishment's dynamic problem would need an additional state variable on employment size, which complicates the analysis. By introducing this assumption, I ignore an additional employment gain of the Hartz reform in hiring more production labor at the same quality level and in hiring more R&D labor with returns to vertical and horizontal innovation increased due to that production margin. Thus, this model is likely to underestimate gains from the Hartz Reform.

at around 1-1.5% in a given year both in the early 2000s and the late 2010s. Thus, I set the baseline c_χ as a very low value, 0.1. In Appendix C.2, I also test various values for c_χ, but the results are similar regardless of the entry elasticity because the entrant innovation does not contribute much to the aggregate growth.

5.2 Steady-State Results

Table 4 compares the steady-state moments before and after the Hartz reform. The first column summarizes the BGP before the Hartz reform, and the fourth column summarizes the BGP after the Hartz reform. To separate the effect of an exogenous increase in labor supply with the effect of removing firing costs, the second column summarizes the steady state after removing the labor wedge without reducing κ. Similarly, the third column fixes the positive labor wedge and only reduces κ to zero.

The labor wedge margin is more important in accelerating productivity growth in the steady state than the firing cost margin. This is because, while the increase in labor supply decreases the wage paid by establishments, the reduction of firing cost increases the wage level. This wage increase after removing firing costs comes from more labor demand for R&D. However, without a further increase in labor supply, innovation rates cannot increase significantly. The decrease in wage after removing the labor wedge is important in reducing innovation costs and boosting incumbents' innovation behavior.

Reducing firing costs contributes to a higher frequency of vertical, horizontal, and entrant innovation, but the impact on growth is slightly negative because the average innovation step size of horizontal and entrant innovation decreases. Successful creative destruction by horizontally expanding incumbents improves the relative quality of a product line on average by 7.47% before the reform. This average step-size becomes 7.23% after the reform. This smaller innovation step size is because the cross-sectional standard deviation of relative quality over x decreases from 12.43% to 12.05% with relatively more horizontal innovation

after removing κ. Removing κ increases horizontal innovation relative to vertical innovation. Since successful horizontal innovation makes the quality of a product line catch up with the better quality level of the expanding establishment, the dispersion of x tends to decrease with more horizontal innovation. The average gain from successful horizontal innovation and switching to a better quality becomes smaller with lower cross-sectional dispersion in the quality space.

Despite the negative effects of removing firing costs, the reform overall increases vertical and horizontal innovation rates per product line, though the increase is more pronounced for vertical innovation. The average vertical innovation rate increases 0.86 percentage points per year, while the average horizontal innovation rate increases only by 0.01 percentage points. Moreover, the job reallocation rates barely change across the different steady states because only vertical innovation has a noticeable response to the Hartz reform.

Table 4: Effect of Hartz Reform in the Steady State

Moment	Pre	No Labor Wedge	No Firing Cost	Post
Output Growth Rate	1.23	1.30	1.23	1.30
Wage	1.097	0.999	1.099	1.000
R&D Labor Share	29.2	27.4	29.2	27.4
Job Reallocation	1.73	1.73	1.73	1.73
Vertical Innovation	16.1	16.9	16.1	16.9
Horizontal Innovation	0.52	0.53	0.52	0.53

Note: All in percent, except Wage. All flow rates are annual. The innovation rates are defined per product line and calculated conditional on successful innovation. Thus, q^v and q^h are multiplied with $\xi(x)$ and $\zeta(x)$, respectively, before taking average over the distribution of x. The job reallocation rate is defined in Section 3.4.

Table 5: Growth and Reallocation Decomposition under the Hartz Reform

Moment	Pre	No Labor Wedge	No Firing Cost	Post
Job Creation from Vertical Innovation	25.4	26.2	25.5	26.2
Job Creation from Horizontal Innovation	26.8	26.8	26.8	26.8
Job Creation Due Entrant Innovation	47.8	47.0	47.8	47.0
Job Destruction from Creative Destruction	12.5	13.1	12.5	13.1
Job Destruction from Obsolescence	29.8	29.1	29.8	29.0
Job Destruction from Exit	57.7	57.8	57.7	57.8
Growth from Vertical Innovation	91.3	90.5	91.8	90.8
Growth from Horizontal Innovation	2.6	2.9	2.5	2.8
Growth from Entrant Innovation	6.1	6.7	5.8	6.4

Note: The job creation and destruction are defined at the product line level. Specifically, job creation happens after vertical, horizontal, and entrant innovation. Job creation by each innovation is the frequencies of innovation multiplied by the increments in production and R&D labor hired by the innovated product line after each innovation. Job destruction happens with quality depreciation, creative destruction, and exogenous exits. Job destruction is calculated similarly by multiplying the event rate by the job loss conditional on the event. The growth rate decomposition formula is described in Section 3.7.

Table 5 shows the contribution of different types of innovation to aggregate growth and labor reallocation. Vertical innovation accounts for most productivity growth, more than 90%. However, vertical innovation accounts for only 23-25% of job creation. This is because vertical innovation does not replace workers within establishments but only adds more workers on top of existing product lines. On the other hand, relative to its contribution, horizontal innovation generates more job creation in the economy with worker replacements across establishments. Entrant innovation also creates more jobs relative to its contribution to growth. Most job destruction comes from exogenous exits. Still, deprecation of relative quality contributes more to destruction than direct creative destruction.

After the full Hartz reform, the contribution of vertical innovation to growth decreases because, in relative terms, horizontal innovation increases more than vertical innovation.

However, because horizontal innovation has such a small share in the growth process from the beginning, the increase in horizontal innovation does not amount to a notable difference after the reform. Entrant innovation also increases its contribution to growth in relative terms, but its absolute impact is minimal. Unlike productivity growth, the contribution of vertical innovation on job creation increases after the reform, largely because demand for R&D labor increases more after vertical innovation. The decrease in wage after removing the labor wedge further contributes to more increases in production labor demand after vertical innovation.

5.3 Transition Path Analysis

In the transition path, I assume an exponentially decreasing labor wedge and firing costs after the unexpected policy introduction at $t = 0$. Motivated by the fact that the reform (Hartz I, II, III, and IV) was implemented over three years, I set the exponent to decrease linearly so that the values of the labor wedge and the firing cost parameters at $t = 36$ months become 1% of the pre-reform level. This gives me 12% as the decaying rate for a unit time interval. Additionally, I set the initial labor wedge and the firing cost parameters at $t = 0$ as $\exp(-0.12) = 88\%$ of the steady-state level. Figure 8 describes the resulting path for labor wedge and firing cost parameters.

Figure 8: Labor Wedge and Firing Cost under the Hartz Reform Transition

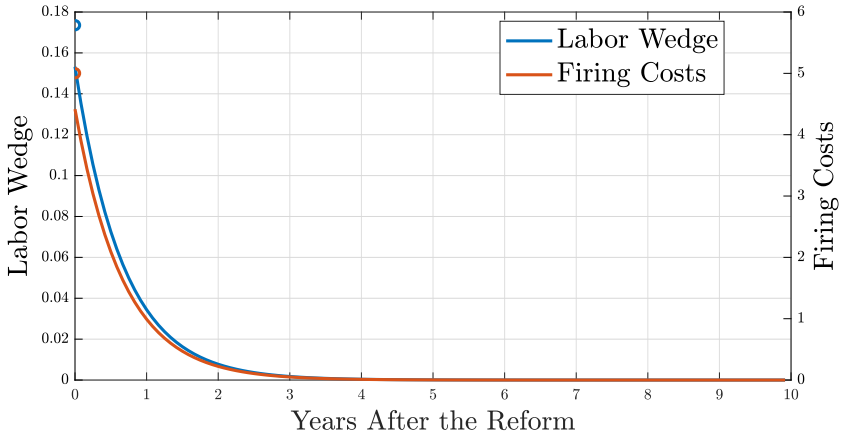


Figure 9 shows the increase in output relative to the pre-reform BGP. The decrease in labor wedge is quantitatively more important than the decrease in firing costs. With the increases in labor supply, the output jumps up initially after the reform in time 0 and grows further with the decaying labor wedge. Moreover, it also grows faster than the pre-reform BGP because of more innovation (see Figure 10). On the transition path, it is also the vertical innovation that contributes the most to output growth. Horizontal innovation and entrant innovation have a positive but limited role in promoting output growth.

Without an exogenous increase in labor supply, the economy experiences an initial drop in output production (as shown by the yellow line in Figure 9) because more labor inputs are used for R&D. Moreover, the economy experiences a permanent output drop after removing the firing costs. This permanent reduction in output results from more innovations that consume labor for R&D but fail to improve productivity growth. As mentioned in the steady-state comparison, horizontal and entrant innovations become less impactful with smaller cross-sectional quality dispersion. Therefore, the economy is worse off by only removing firing costs. This negative output effect also happens after removing firing costs for the same reduction in the labor wedge.

Figure 9: Relative Log Output under the Hartz Reform Transition

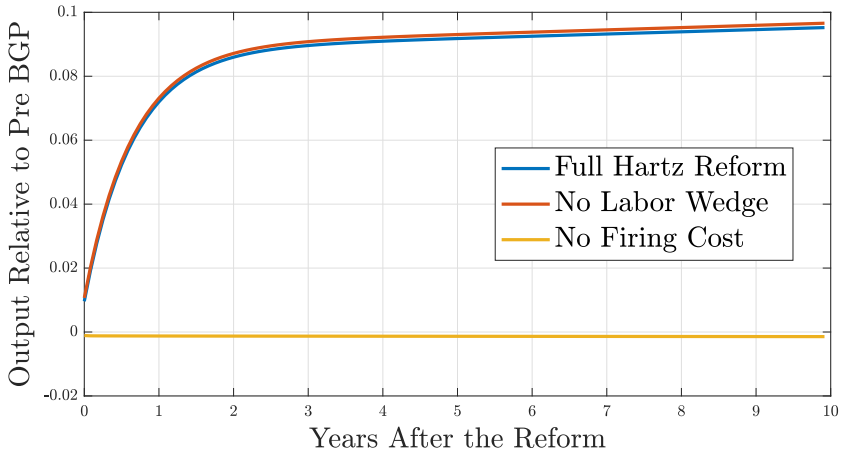


Figure 10: Relative Contribution to Output under the Hartz Reform Transition

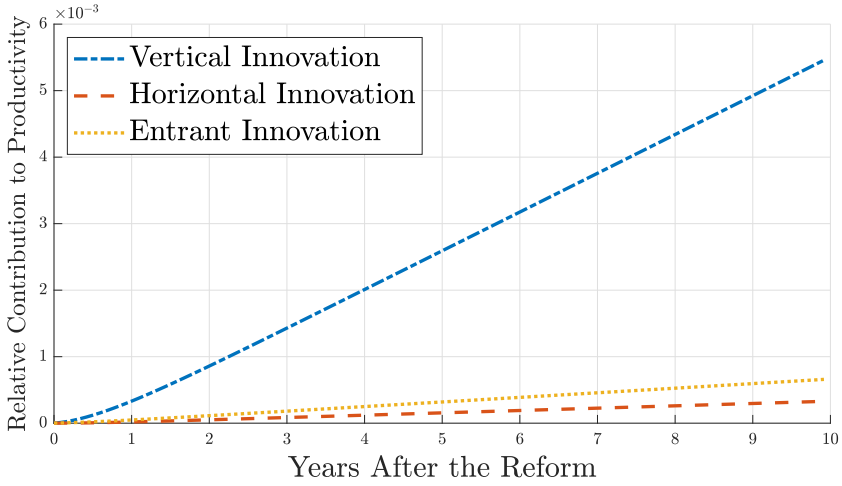
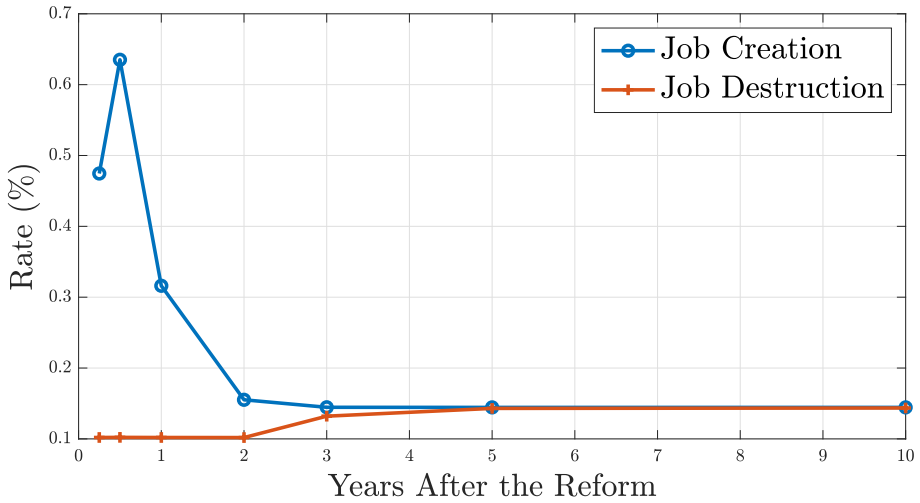


Figure 11 describes the changes in job creation and destruction rates (for derivations of job creation and destruction rates over the transition path, see Appendix C.1). Because of a sharp reduction in labor wedge and wage at $t = 6$ and $t = 12$, even non-innovating establishments contribute to job creation in the short run after the reform. Over the transition path, only establishments with the exit or creative destruction shots make job destruction. Consequently, in the short run after the reform, job creation increases but job destruction

decreases. However, four years after the reform, the economy still has higher job creation rates but lower job destruction rates than the pre-reform steady state.

Figure 11: Job Creation and Destruction Rates over the Transition Path



Note: Job creation and destruction rates over the transition path are defined in Appendix C.1. $\Delta = 1$.

The third column of Table 6 shows that the welfare effect of Hartz reform is equivalent to a permanent 6.1% increase in the consumption schedule of the representative household in the pre-reform BGP. Output growth from more vertical innovation is the most important factor of the welfare increase in this case. The dynamic gains from faster output growth after the Hartz reform accounts for 17% (0.98/5.87) of the entire increase in welfare, while the remainder originates from the static increase in production labor right after the reform. Vertical innovation accounts for 85% (0.83/0.98) of dynamic gains from the reform. Consistent with the output results over the transition path, reducing labor wedge plays a dominant role in improving welfare. Without increases in labor supply, the reform that removes firing costs reduces the welfare as additional R&D activities soak up production labor and even slow down productivity growth in the future. Again, this is because the average step sizes of horizontal and entrant innovation decrease with smaller cross-sectional dispersion with more

horizontal innovation.

Table 6: Effect of the Hartz Reform in the Transition Path

	No Labor Wedge	No Firing Cost	Post
Consumption Equivalent Variation	6.05	-0.18	5.87
Initial Output Change	1.07	-0.11	0.97
Vertical Innovation	0.80	0.05	0.83
Horizontal Innovation	0.06	-0.02	0.05
Entrant Innovation	0.15	-0.06	0.10
Labor Supply	-3.22	-0.04	-3.27

Note: All in percent. The Consumption Equivalent Variation is a permanent increase in consumption for the representative household in the baseline economy to have the same lifetime utility as in the transition economy. Vertical, Horizontal, and Entrant Innovations are the contributions of each innovation on the welfare change. Labor Supply is the change in welfare from increased labor supply. The rest of welfare change comes from additional production labor over the transition path. For the formal definitions, see Appendix C.1.

In Appendix C.4, I iterate the numerical exercise without vertical innovation in the model. I calibrate the model without vertical innovation and repeat the Hartz counterfactual analysis. The model without vertical innovation predicts smaller increases in growth rates and welfare. More importantly, job creation and destruction rates respond more elastically to the reform even with a smaller increase in growth rates.

6 Concluding Remarks

This paper is one of the first papers that use the surveyed measure of establishment-level innovation activities and relate the measure to establishment performance after the innovation events and output growth in the general equilibrium. This measure covers knowledge creation and adoption activities in broader sectors, such as service sectors, than the patent-based measures. Moreover, the survey-based measures in the German panel make it

possible to distinguish vertical and horizontal innovation. Using this new measure, this paper establishes new empirical facts on the characteristics of innovating establishments before and after innovation events. Moreover, this paper finds suggestive evidence about equilibrium responses of non-innovating establishments to more frequent innovation activities of competitors in the same sector.

This paper constructs a model where two innovation choices by incumbent establishments can affect endogenous productivity growth. The model is simple yet able to accommodate the new facts from the survey-based measure of innovation. The calibrated model is then used to quantify the macroeconomic effects of a major labor market reform in Germany through endogenous innovation choices. The model predicts that the welfare of the overall German economy improves with more production labor right after the Hartz reform and is further amplified through faster productivity growth. Productivity growth accounts for 17% of the entire welfare gains from the reform, where vertical innovation explains 85% of the gains from productivity growth. Since only vertical innovation responds significantly to the reform, despite the growth responses, the labor turnover barely changes in the steady state and a few years after the reform.

This paper further emphasizes the importance of vertical innovation in decoupling the co-movement between productivity growth and job turnover. Unlike horizontal innovation, vertical innovation raises productivity even without large labor adjustments. The main idea in this paper may provide an explanation on why developed countries with slower business dynamism could maintain stable output growth, which is left for future research.

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A Data Appendix

A.1 Innovation Measures

The IAB-BP asks its respondents whether they had the following four types of innovation activities in the previous year of the survey. Two of the questions are used as a measure of vertical innovation, and one of them is used as a measure of horizontal innovation. The other question is not used in the paper but enters as a control in the regression equation in A.3.

1. **Quality Improvement**: In the last business year, did your establishment improve or further develop a product or service which had previously been part of your portfolio? ($\in$ vertical innovation in the paper)

2. **Product Expansion**: In the last business year, did your establishment start to offer a product/service that had been available on the market before? (= horizontal innovation in the paper)

3. **Variety Creation**: Have you started to offer a completely new product or service in the last year for which a new market had to be created? (excluded from the paper)

4. **Process Innovation**: Did you develop or implement procedures in the last business year which have noticeably improved production processes or services? ($\in$ vertical innovation in the paper)

Table A.1 shows the frequencies of innovation separately for four different questions. The number of establishments is higher for bins with larger establishments because smaller establishments tend to be under-represented in the overall sampling procedure. Therefore, all regression analyses below use these cross-sectional weights from the data provider in the study.

Table A.1: Raw Innovation Frequencies across Size (in %)

	Quality Improv.	Product Expansion	New Variety	Process Innov.	Any Innov.	N. Est.
All	31.01	20.09	5.76	11.77	39.70	91,823
Small	23.01	14.82	3.36	7.09	30.72	10,363
Medium	27.18	16.80	3.80	9.20	34.78	11,263
Large	38.60	23.56	7.95	16.61	47.94	42,670

Table A.2 shows the innovation frequencies across equal-sized establishment age groups. As reported in previous studies using patent measures, such as Acemoglu et al. (2018), younger establishments are more likely to innovate both horizontally and vertically, despite the positive correlation between size and age. However, the age dependency is flatter than the size dependency.

Table A.2: Innovation Frequencies across Age (in %)

	Quality Improv.	Product Expansion	New Variety	Process Innov.	Any Innov.	N. Est.
All	31.01	20.09	5.76	11.77	39.70	91,823
Young	32.50	23.00	6.85	12.50	42.08	23,954
Medium	30.61	18.57	5.49	11.18	38.74	26,306
Old	29.87	18.43	4.92	11.49	38.14	41,563

Figures A.1 and A.2 display the distribution of the numbers of vertically and horizontally innovation years for each establishment, respectively. Since innovation questions appear in ten years of the survey, the maximum number of innovation years is ten. Innovation is concentrated around a small subset of establishments in the survey. Some establishments do not engage in innovation activities at all while a few establishments do innovate almost every year. Overall, the distribution for vertical innovation is placed to the right of the distribution for horizontal innovation.

Figure A.1: Number of Vertically Innovation Year Per Est.

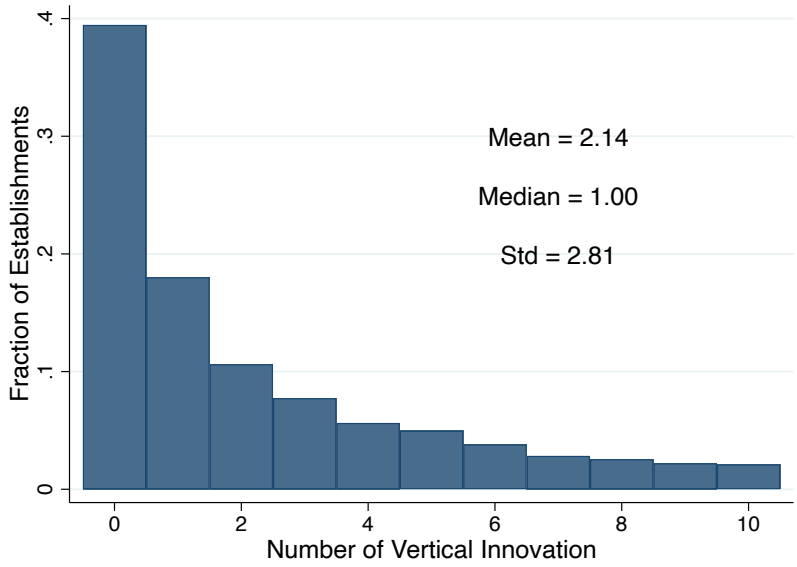
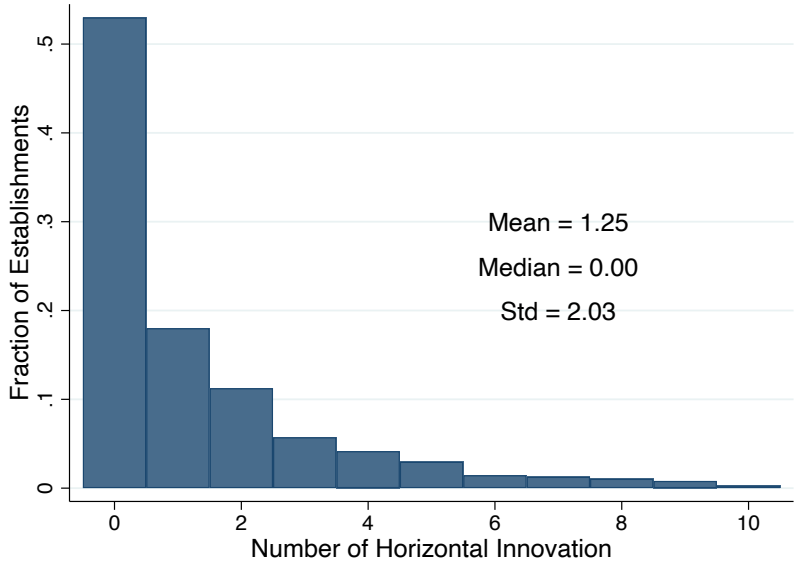


Figure A.2: Number of Horizontally Innovation Year Per Est.



Because of this high heterogeneity across establishments in Figures A.1 and A.2, inno-

vation in a given year predicts a higher probability of innovation in other years in Figures A.3 and A.4. First, both vertical and horizontal innovations have a positive auto-correlation. Vertical and horizontal innovations in year 0 predict higher probabilities of vertical and horizontal innovations in other considered years relative to the averages, indicated by the navy dashed lines for vertical innovation and the maroon dashed lines for horizontal innovation. Second, vertical and horizontal innovations are positively correlated. Vertical innovation in year 0 is associated with overall higher probabilities of horizontal innovation in all considered periods, and vice versa. Lastly, the forward correlation is smaller than the backward correlation. An innovation in year t predicts a higher probability of innovating before than after. It is consistent with the fraction of innovating establishments decreasing in establishment age in Table A.2.

Figure A.3: Fraction of Innovating Est. Conditional on Vertical Innovation

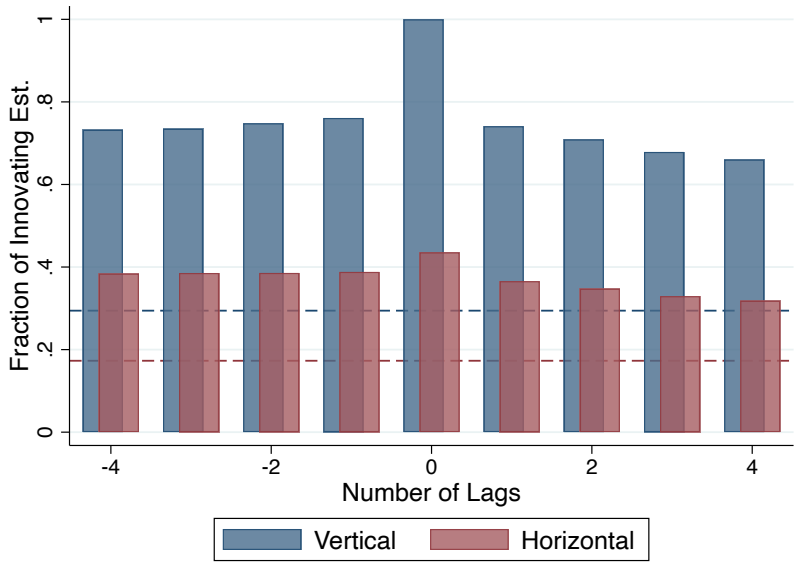
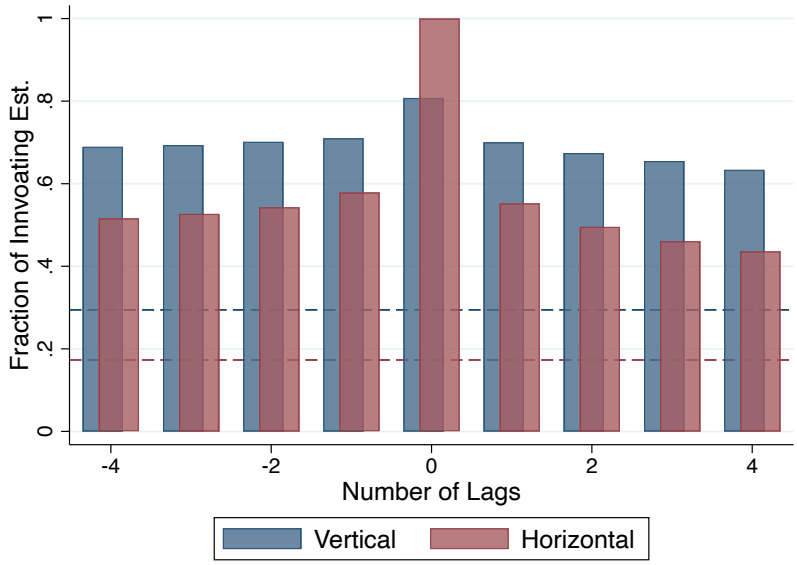


Figure A.4: Fraction of Innovating Est. Conditional on Horizontal Innovation



A.2 Comparison to Patent-Based Measures

I compare the survey-based measured of innovation to the patent-based measures, which have been widely used in innovation literature. Unfortunately, it is impossible to match patents to establishments in IAB-BP, because of the anonymization. As a result, I match the class of patents to the sector of establishment and construct sector × year. I then count the number of patents granted to German entities by the United States Patent and Trademark Office (USPTO) in each sector, using the raw patent-level data parsed and managed by `patentsview.org`. I first translate the cooperative patent class endowed by the USPTO to the 3-digit North America Industry Classification System (NAICS) Code of 2007 using probabilistic crosswalks constructed in Goldschlag et al. (2019). I manually match the 3-digit NAICS code to the sectoral specification of IAB-BP.

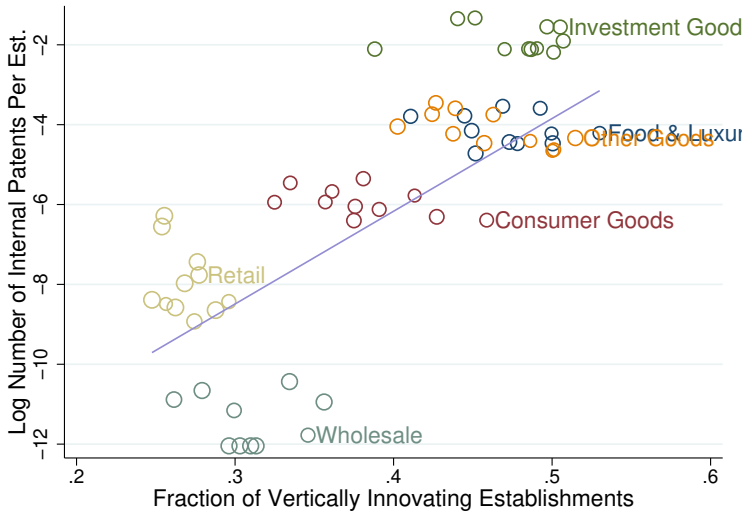
The patent-based measure of innovation does not exist for agriculture, mining, utility, or more importantly, service sectors. Patent measures exist primarily for manufacturing

sectors where measuring new knowledge is more tangible. Innovation made in the service sector is often hard to be patented. For example, software is not protected by patent but rather by copyright. Software patents do exist, but they often serve to drive out their competitors rather than measure actual innovation.

Then, I compare the total number of patents to the fraction of either vertically or horizontally innovating establishments. In Figure A.5, the fraction of innovating establishments in each sector and year is overall positively correlated with the number of patents per establishments in that sector and year in the pooled sample, and the correlation coefficient is 0.55. This positive correlation, however, comes more from sectoral heterogeneities than time-series variations within each sector. Indeed, within each sector, the log number of patents per establishments is almost flat over the fraction of innovating establishments. Incorporating sectoral heterogeneities is preferred in this case because there can be lags between the appearance of frontier knowledge, which is proxied by the patent data, and the adoption of new knowledge, which appears as the fraction of innovating establishments.

Figure A.5: Fraction of Innovating Establishments and Number of Patents

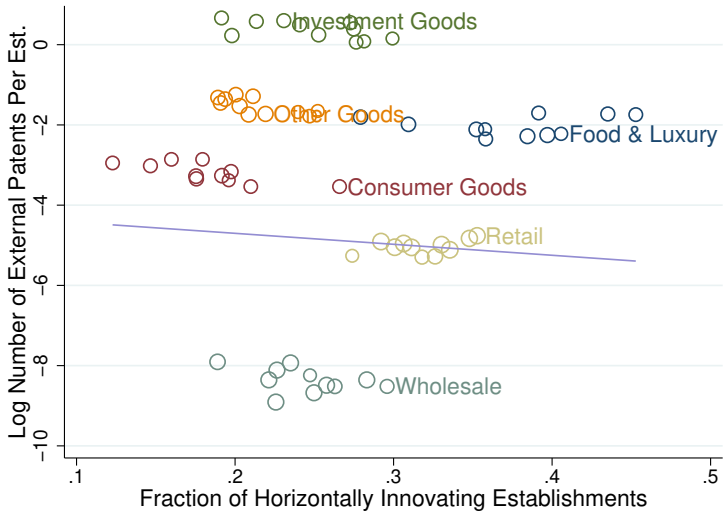
Figure A.6: Fraction of Vertically Innovating Establishments and Number of Internally Patents



In the next two figures, I distinguish vertical and horizontal innovation in the survey measure and external and internal innovation in the patent measure. Akcigit and Kerr (2018) defines internal patents as patents citing patents granted to their assignees for more than half of the total citations. External patents are all patents other than internal patents. Akcigit and Kerr (2018) use internal versus external patents to measure innovation over firms' previous product lines versus innovation made over others' product lines. I classify patents using the previously written criterion and recalculate the number of internal and external patents in each sector and year. Then I draw scatter plots of the number of internal and external patents per establishment against the fraction of vertically and horizontally innovating establishments, respectively.

Figures A.6 and A.7 show the results. In Figure A.6, the vertical innovation measure at the sector and year level is positively associated with the number of internal patents per establishment. This positive correlation again comes more from the sectoral heterogeneities in the average fraction of vertical innovation and the number of internal patents than from

Figure A.7: Fraction of Horizontally Innovating Establishments and Number of Externally Patents



within-sector variations. Meanwhile, in Figure A.7, the measure of horizontal innovation is almost uncorrelated with the number of external patents per establishment. The retail sector has a relatively higher fraction of horizontally innovating establishments than other manufacturing sectors, but its number of external patents is lower than the capital good sector. This contrasts with the vertical innovation case where the retail sector has both a lower fraction of vertically innovating establishments than most manufacturing establishments and a lower number of vertical patents per establishment.

A.3 Surveyed Innovation and Establishment Performance

To investigate the implications of innovation measures at the establishment level more rigorously, I regress a statistical model of dynamic correlations of establishment-level behaviors with innovation dummies using the specification below. First, the dependent variables on growth are defined in a way similar to Davis et al. (1998).

- Employment growth rate $= (AdmEmp - L.AdmEmp)/(\frac{1}{2}AdmEmp + \frac{1}{2}L.AdmEmp)$

- Hiring rate $= 2 \times Hiring/(\frac{1}{2}AdmEmp + \frac{1}{2}L.AdmEmp)$
- Separation rate $= 2 \times Separation/(\frac{1}{2}AdmEmp + \frac{1}{2}L.AdmEmp)$
- Wage, Revenue, Investment growth rates are defined similarly
- Hiring rate $= 2 \times Hiring/(\frac{1}{2}AdmEmp + \frac{1}{2}L.AdmEmp)$
- Separation rate $= 2 \times Separation/(\frac{1}{2}AdmEmp + \frac{1}{2}L.AdmEmp)$
- Fraction of Trained Employees $= 2 \times \#ClassParticipation/(\frac{1}{2}AdmEmp + \frac{1}{2}L.AdmEmp)$
- Establishment age = Survey Year - Year of Foundation,

where

- $AdmEmp$ = Employment of an establishment as of June 30 in the survey year from the administrative data
- $Hiring$ = Total number of newly hired workers in the first half of the survey year from the survey
- $Separation$ = Total number of separated workers in the first half of the survey year from the survey
- $Wage\,expense$ = gross wage payment as of June in the survey year from the administrative data
- $Revenue$ = Business volume in the previous year from the survey
- $Investment$ = Sum of investment expenditures in the previous year from the survey
- $\#ClassParticipation$ = The number of classes participations at least partially covered by the establishment from the survey

All samples are weighted by the cross-sectional weight provided by the IAB. Since the year of establishment is left-censored in 1990, I replace the year of establishment with the year of

1990 if the establishment existed before 1990. I exclude observations with missing age. Table A.3 shows the summary statistics for the dependent variables in the regression exercise.

Table A.3: Summary Statistics - Outcome Variables (in %)

	Mean	Sd	Median	1st Q.	3rd Q.	N.
Employment Gr.	0.72	22.06	0.00	0.00	4.88	110,946
Hiring Rate	13.77	40.56	0.00	0.00	10.53	99,046
Separation Rate	11.39	36.18	0.00	0.00	6.30	99,112
Wage Gr.	2.14	14.65	1.27	-1.52	5.93	91,358
Rev. Gr.	2.13	27.86	0.00	-7.68	12.39	77,919
Invest. Gr.	-5.63	1.55	0.00	-198.01	169.23	81,008
Training Rate	79.73	184.23	24.29	0.00	131.58	85,546

Overall employment-related variables imply that most of the establishments in the sample neither hire new workers nor separate from their existing workers. As a result, they stay at their initial employment size while an average establishment grows *conditional on survival*. Wage and revenue increase, but investment dwindles on average as the establishment ages. An average establishment has approximately 80% of workers trained, but a median firm has only 24.3% of its employees trained per year (at least at a partial expense of the establishment[30]). Finally, establishments show very heterogeneous dynamics in employment-related variables.

Then, the following regression specifications similar to event studies estimate dynamic correlations of innovation dummies with dependent variables before and after the innovation events.

$$Y_{it} = \sum_{s=-5}^{5} \mathbf{1}^{a}_{it+s} \beta^{a}_{s} + \delta_i + \gamma_t + \varepsilon_{it} \tag{A.1}$$

[30]Since this question asks the number of class participation in the training program, it can be larger than the number of employees.

Y_{it} is employment growth, hiring, or separation rates of establishment i at year t, $\mathbf{1}_{is}^{a}$ is indicator of innovation type a, either vertical innovation ($verti$) or horizontal innovation ($horiz$). δ_i is the establishment fixed effect, and γ_t is the year fixed effect.

In this regression, I use all samples from 2003 - five years before the beginning of the innovation modules - to estimate the pre-trend. Since the survey asks about innovation activities in the previous year, while the wage expense and the employment variables are defined as of June of the survey year, the growth rate after the innovation may be the one between the innovation year and the previous year of the innovation. However, as shown in Section 2.2, most firms do not experience a size growth relative to an average firm between the innovation year and the previous year. Thus, unlike in the Figure 3, I set the employment growth rate between the survey year and the previous year as the growth rate of the innovation year ($s = 0$) if the innovation variable takes the value of one in the same survey. Similarly, because the numbers of hiring and separation are defined in the first six months of the survey year, the hiring and separation rates of the current year are defined as the hiring and separation rates of the innovation year if the innovation activities of the previous year took place in the survey of the current year. The lagged terms in the regression equations are defined in the regression accordingly. The profit and the investment are surveyed as being made in the last year. Likewise, the profit and the investment of the survey year are the profit and the investment of the innovation year if the innovation dummies are one in the survey of the current year.

I further consider the following specification where I include all the innovation types surveyed in the data. This specification controls for any spurious correlation of individual

innovation types to outcome variables Y_{it} through other types of innovation[31].

$$Y_{it} = \sum_{s=-5}^{5} \mathbf{1}_{it+s}^{verti} \beta_s^{verti} + \sum_{s=-5}^{5} \mathbf{1}_{it+s}^{horiz} \beta_s^{horiz} + \sum_{s=-5}^{5} \mathbf{1}_{it+s}^{newvar} \beta_s^{newvar} + \delta_i + \gamma_t + \varepsilon_{it} \tag{A.2}$$

Here, $a = newvar$ indicates the creation of entirely new products, which is also reported in the survey.

Table A.4 shows estimation results on employment-related variables. Vertical innovation predicts an increase in employment growth by 1.05 percentage points in the innovation year, but the effect disappears three years after the vertical innovation. On the other hand, horizontal innovation correlates significantly with increases in employment in the innovation year. This correlation accumulates over time and becomes larger four years (Compare the third to the sixth columns.). The employment growth after vertical and horizontal innovation entirely comes from additional hiring. Notably, horizontal innovation increases hiring rates by 2.3 percentage points in the innovation year in the third column of the second panel in Table A.4. Due to large measurement errors associated with the surveyed number of new hires, the coefficient of horizontal innovation becomes less significant but still stays high three years after the innovation. The coefficients of vertical and horizontal innovation on separation rates are overall insignificant and small in magnitude. The only exception, vertical innovation over four-year lags, indicates that the vertical innovation may be associated with an *increase* in the separation rates cumulatively three years after the vertical innovation. This is consistent with the fact that vertical innovation features an initial spike in employment but predicts no impact three years after.

[31] Samples with missing growth rates are excluded. These samples are often entrant and exiting establishments but also contain samples that disappeared or are added in the panel.

Table A.4: Regression Results

Dependent Variable: Employment Growth Rates

N. Lags	1 Year	1 Year	1 Year	4 Years	4 Years	4 Years
Vertical Innov.	1.05***		0.85***	-0.34		-0.93
	(0.31)		(0.32)	(0.67)		(0.75)
Horizontal Innov.		1.23***	1.10***		2.42***	3.25***
		(0.35)	(0.38)		(0.83)	(0.97)
N.	100,770	100,770	100,770	100,770	100,770	100,770
Other Innov.			✓			✓
Fixed Effects	Est.	Est.	Est.	Est.	Est.	Est.

Dependent Variable: Hiring Rates

N. Lags	1 Year	1 Year	1 Year	4 Years	4 Years	4 Years
Vertical Innov.	1.98***		1.26**	2.33*		0.97
	(0.59)		(0.62)	(1.28)		(1.44)
Horizontal Innov.		2.75***	2.41***		3.80**	3.34*
		(0.68)	(0.73)		(1.60)	(1.86)
N.	100,406	100,406	100,406	100,406	100,406	100,406
Other Innov.			✓			✓
Fixed Effects	Est.	Est.	Est.	Est.	Est.	Est.

Dependent Variable: Separation Rates

N. Lags	1 Year	1 Year	1 Year	4 Years	4 Years	4 Years
Vertical Innov.	0.56		0.28	2.96**		2.61*
	(0.59)		(0.62)	(1.28)		(1.45)
Horizontal Innov.		0.76	0.62		1.28	-0.09
		(0.68)	(0.73)		(1.60)	(1.86)
N.	100,475	100,475	100,475	100,475	100,475	100,475
Other Innov.			✓			✓
Fixed Effects	Est.	Est.	Est.	Est.	Est.	Est.

Note: In the columns specified as 1 Year and 4 Year, estimates for β_0^a and $\sum_{s=0}^{3} \beta_s^a$ are reported, respectively. The third and the sixth columns follow regression equation (2), and all the other columns follow regression specification (1). Numbers in the parentheses are the standard errors of the estimates. $***$ - Significant at 1%, $**$ - Significant at 5%, $*$ - Significant at 10%. All samples are weighted by the cross-sectional weight from the data provider.

These properties are incorporated into the structural model of endogenous innovation

choice in Section 3. In the model, vertical innovation is associated with an increase in the quality of a product line by an establishment originally holding the line. This leads to a small increase in size, and the effect disappears quickly as the quality depreciates over time. On the other hand, horizontal innovation is modeled as an expansion of an establishment to a new product line. As a result, employment size jumps up by a larger amount, and the effect lasts until the new product line is taken by another establishment.

More specifically in the model, the step size of vertical innovation is constant regardless of the previous productivity level, which implies constant production labor growth after vertical innovation. The rationale for this assumption is that the growth rate of size after vertical innovation is comparable and not significantly different between small and medium groups, which are more likely to have only one product line. The size growth in the vertical innovation year is 2.32% for the small group and 2.61% for the medium group[32]. They are similar in magnitude and not significantly different from each other.

On the other hand, horizontal innovation is assumed as an additional product line of which the productivity is the same as the underlying product line. This assumption contrasts with assumptions made in other Schumpeterian models such as Akcigit and Kerr (2018) where the step size distribution is the same for all external innovation. This assumption is to replicate the horizontal innovation rate which is increasing in size even between the small and the medium group. Establishment size is positively correlated with underlying productivity. Thus, the innovation frequency that is increasing in size is interpreted as the innovation frequency that is increasing in underlying productivity. Another rationale of this assumption is that the imputed mean wage of full-time employees from the administrative data set does not substantially and significantly increase after the horizontal innovation, unlike vertical innovation, regardless of the lagged size group of the establish-

[32]The 'small' group consists of establishments in the first three deciles of the size distribution, while the 'medium' group ones are the establishments in the second three deciles of the employment size distribution.

ments. The magnitude of the wage increase is also close to zero. As long as productivity growth is reflected in wages, this result implies that horizontal innovation does not increase the productivity of the establishment.

In Table A.5, I show the regression results of other dependent variables on wage, revenue, investment growth rates, and the fraction of trained employees. Vertical innovation is associated with wage and revenue increases in the innovation year but leads to decreases in wage and revenue three years after. This, as in the case of employment, suggests that the increase in profitability after vertical innovation disappears quickly and the establishment eventually shrinks without further innovation. Horizontal innovation is not associated with wage increases, but it leads to a larger increase in revenue. Due to substantial measurement errors associated with survey measures on revenue, overall standard errors are larger than the variables from the administrative data set. As a result, the significance of horizontal innovation disappears with a large standard error with four years of lags, but the magnitude remains quite high even three years after the innovation year. This result is consistent with the interpretation that the new product opportunity from horizontal innovation gives a larger increase in revenue and lasts longer than vertical innovation. Both innovation types are associated with a large increase in investment. Lastly, horizontal innovation predicts that more employees are trained at the establishment's expense.

Table A.5: Regression Results

Dependent Variable: Wage Growth Rates

N. Lags	1 Year	1 Year	1 Year	4 Years	4 Years	4 Years
Vertical Innov.	0.60***		0.65***	-0.79**		-0.89**
	(0.17)		(0.18)	(0.37)		(0.41)
Horizontal Innov.		-0.10	-0.31		-0.18	0.05
		(0.20)	(0.21)		(0.46)	(0.53)
N.	91,482	91,482	91,482	91,482	91,482	91,482
Other Innov.			✓			✓
Fixed Effects	Est.	Est.	Est.	Est.	Est.	Est.

Dependent Variable: Revenue Growth Rates

N. Lags	1 Year	1 Year	1 Year	4 Years	4 Years	4 Years
Vertical Innov.	1.33***		0.98**	-2.34***		-2.70***
	(0.36)		(0.38)	(0.78)		(0.87)
Horizontal Innov.		1.27***	0.86**		0.36	1.12
		(0.41)	(0.44)		(0.97)	(1.12)
N.	71,318	71,318	71,318	71,318	71,318	71,318
Other Innov.			✓			✓
Fixed Effects	Est.	Est.	Est.	Est.	Est.	Est.

Dependent Variable: Investment Growth Rates

N. Lags	1 Year	1 Year	1 Year	4 Years	4 Years	4 Years
Vertical Innov.	29.06***		24.88***	12.37***		8.49*
	(2.09)		(2.20)	(4.53)		(5.09)
Horizontal Innov.		25.00***	15.64***		18.51***	12.77*
		(2.38)	(2.55)		(5.66)	(6.54)
N.	73,851	73,851	73,851	73,851	73,851	73,851
Other Innov.			✓			✓
Fixed Effects	Est.	Est.	Est.	Est.	Est.	Est.

Dependent Variable: Fraction of Trained Employees

N. Lags	1 Year	1 Year	1 Year	4 Years	4 Years	4 Years
Vertical Innov.	26.89***		15.92*	36.22**		-0.37
	(7.88)		(8.36)	(18.21)		(20.66)
Horizontal Innov.		37.80***	27.49***		74.46***	48.87*
		(9.03)	(9.78)		(22.71)	(26.56)
N.	86,869	86,869	86,869	86,869	86,869	86,869
Other Innov.			✓			✓
Fixed Effects	Est.	Est.	Est.	Est.	Est.	Est.

Note: In the columns specified as 1 Year and 4 Year, estimates for β_0^a and $\sum_{s=0}^{3} \beta_s^a$ are reported, respectively. The third and the sixth columns follow regression equation (2), and all the other columns follow regression specification (1). Numbers in the parentheses are the standard errors of the estimates. $***$ - Significant at 1%, $**$ - Significant at 5%, $*$ - Significant at 10%.

A.4 Sector-level Innovation and Non-Innovating Establishments

Another prediction of Schumpeterian models is that innovating establishments replace non-innovating establishments in the market, also known as 'creative destruction'. I define the market as the composition of establishments in the same year and sector and investigate how average innovation behavior in a market is associated with the behavior of non-innovating establishments. Then, the following establishment-level regression equations are considered. Note that the samples for the regression only include non-innovating establishments, which do not innovate vertically or horizontally in years t, $t-1$, and $t+1$.

$$Y_{it} = \beta^I I_{j(i)t} + \gamma' X_{it} + \epsilon_{it} \tag{A.3}$$

$$Y_{it} = \beta^V L.V_{j(i)t} + \beta^H L.H_{j(i)t} + \gamma' X_{it} + \epsilon_{it}, \tag{A.4}$$

where Y_{it} is the same dependent variables of establishment i in year t as in Equations A.1 and A.2. $I_{j(i),t}$ is average innovation measure of sector j that i belongs to in time t, and L is the yearly lag operator, $I = V, H$. X_{it} contains controls such as an entrant dummy, the lagged establishment size, the sector-year level Herfindahl-Hirschman index and total employment growth, and fixed effects. The average vertical innovation measure $V_{j(i)t}$ is defined by the weighted fraction of establishments that experienced vertical innovation in sector $j(i)$ in year t. The average horizontal innovation measure $H_{j(i)}$ is determined similarly.
I first consider separate regressions for different types as in Equation A.3. However, because of the positive correlation among innovation measures, I assume additive separability across impacts of different innovation measures on non-innovation establishments' performance and also consider Equation A.4. In this regression, I also include the employment share of entrant establishments in sector $j(i)$ and in year t. This variable is used to proxy entrant innovation.

Table A.6 shows the regression results from estimating Equations A.3 and A.4. In the

first panel, the size growth among non-innovating surviving firms is much lower in a sector of a year with a higher fraction of horizontal innovating establishments. This result is even robust after controlling for the sectoral heterogeneity in size growth and capturing only time-series variations within each sector. Often in the literature, this negative effect is hard to find because business-cycle shocks tend to move all establishments in the same direction regardless of their innovation status. As for vertical innovation, its coefficient is often smaller and insignificant. This supports the idea that horizontal innovation is important in generating job flows.

The second panel of Table A.6 shows the correlation of sector-wise innovation with the extensive margin of establishment dynamics. The outcome variable Y_{it} is now the establishment exit dummy, which takes the value of one only if the establishment existed in the previous year's panel and exits in the next year. Again, the exit behavior, as opposed to sample attrition, is identified by the variable indicating the last year the establishment appears in the administrative data. The exit dummy takes the value of zero if the sample existed in the previous year panel and exists in the market in the current year.

The coefficient estimate for β^H changes its sign after controlling for the sector fixed effects. Before including industry fixed effects, the estimate is significantly negative, consistent with the theory that more horizontal innovation drives out non-innovating establishments in both the extensive margin and the intensive margin. However, after including industry fixed effects, the estimate becomes significantly positive. This is because the underlying shock behind the time series variations within each industry is more likely to have business cycle shock components that affect all establishments in the same direction. The sector-level variation is more likely to catch the long-run effect of persistent heterogeneity of industries over innovation frequencies.

The last panel of Table A.6 shows the correlation between industry-level innovation measures and the wage growth of non-innovating establishments. The coefficient estimate is

significantly positive, implying that the innovation has spillover effects of increasing wages among non-innovating establishments. This may contribute to the smaller size growth of the non-innovating establishment.

Table A.6: Industry-level Innovation and Non-innovating Establishments

	Dep. Variable: Employment Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
Vertical Innov.	-2.03		3.34	-24.17***		-17.35
	(2.57)		(2.92)	(8.89)		(9.28)
Horizontal Innov.		-9.34***	-10.85***		-32.05***	-29.51***
		(2.57)	(2.92)		(9.60)	(10.13)
Entrant Emp. Share			-38.85**			-53.71
			(19.11)			(30.03)
N.	20,997	20,997	20,997	20,839	20,839	20,839
Fixed Effects	Year	Year	Year	Year & Sector	Year & Sector	Year & Sector

	Dep. Variable: Establishment Exit					
	(1)	(2)	(3)	(4)	(5)	(6)
Vertical Innov.	4.00**		1.21	6.11		9.66
	(1.60)		(1.82)	(5.44)		(5.68)
Horizontal Innov.		6.15***	5.49***		-10.34	-12.79**
		(1.60)	(1.82)		(5.89)	(6.21)
Entrant Emp. Share			33.97***			10.06
			(11.91)			(18.41)
N.	21,149	21,149	21,149	20,989	20,989	20,989
Fixed Effects	Year	Year	Year	Year & Sector	Year & Sector	Year & Sector

	Dep. Variable: Wage Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
Vertical Innov.	3.64**		1.04	4.87		1.71
	(1.44)		(1.69)	(5.18)		(5.40)
Horizontal Innov.		5.81***	5.24***		11.97**	9.75
		(1.48)	(1.74)		(5.62)	(5.93)
Entrant Emp. Share			-23.52**			-32.68
			(11.82)			(17.87)
N.	18,024	18,024	18,024	17,878	17,878	17,878
Fixed Effects	Year	Year	Year	Year & Sector	Year & Sector	Year & Sector

B Model Appendix

B.1 Derivation of the Value Function

The entire firm value is characterized by the following equation:

$$
\begin{aligned}
(r+\delta)\,\mathfrak{V}\left(\left\{x_j\right\}_{j=1}^{N_i}, W_t, Y_t\right) = \max_{\{\xi_j,\zeta_j\}\geq 0} \Bigg\{ & \sum_{j=1}^{N_i} Y_t \pi(x_j) - \sum_{j=1}^{N_i} g_x x_j \mathfrak{V}_{t,x_j}\left(\left\{x_{j'}\right\}_{j'=1}^{N_i}, W_t, Y_t\right) \\
& + g_Y Y_t \mathfrak{V}_{Y_t}\left(\left\{x_{j'}\right\}_{j'=1}^{N_i}, W_t, Y_t\right) + g_w W_t \mathfrak{V}_{w_t}\left(\left\{x_{j'}\right\}_{j'=1}^{N_i}, W_t, Y_t\right) \\
& - W_t \sum_j c_\xi\left(\xi_j\right) + \sum_j \xi_j \left(\mathfrak{V}_t\left(\left\{x_{j'}\right\}_{j'\neq j} \cup \left\{x_j + \lambda^i \bar{x}\right\}, W_t, Y_t\right) - \mathfrak{V}_t\left(\left\{x_{j'}\right\}_{j'=1}^{N_i}, W_t, Y_t\right)\right) \\
& - W_t \sum_j c_\zeta\left(\zeta_j\right) + \sum_j \zeta_j \left(p + (1-p)\,F\left(x_j\right)\right) \\
& \quad \times \left(\mathfrak{V}_t\left(\left\{x_{j'}\right\}_{j'=1}^{N_i} \cup \left\{x_{N_i+1} = x_j\right\}, W_t, Y_t\right) - \mathfrak{V}_t\left(\left\{x_{j'}\right\}_{j'=1}^{N_i}, W_t, Y_t\right)\right) \\
& + \sum_j \tau(x_j)\left(\mathfrak{V}_t\left(\left\{x_{j'}\right\}_{j'=1}^{N_i} \setminus \left\{x_j\right\}, W_t, Y_t\right) - \mathfrak{V}_t\left(\left\{x_{j'}\right\}_{j'=1}^{N_i}, W_t, Y_t\right) - \kappa\right) - W_t \delta \kappa \#\left(\left\{x_j\right\}_{j=1}^{N_i}\right) \Bigg\}
\end{aligned}
$$

where $\mathfrak{V}_{t,x_j}(\left\{x_{j'}\right\}_{j'=1}^{N_i}, W_t, Y_t) = d\mathfrak{V}(\left\{x_{j'}\right\}_{j'=1}^{N_i}, W_t, Y_t)/dx_j$.
First, guess and verify that $\mathfrak{V}_t\left(\left\{x_j\right\}_{j=1}^{N_i}, w_t, Y_t\right) = \sum_{j=1}^{N_i} V_t\left(x_j, w_t, Y_t\right)$, where

$$
\begin{aligned}
\left(r+\delta+\tau\left(x\right)\right) V_t\left(x, W_t, Y_t\right) = \max_{\xi,\zeta\geq 0} \Bigg\{ & Y_t \pi\left(x\right) - g_x x V_t'\left(x, W_t, Y_t\right) \\
& + g_Y Y_t V_{t,Y}(x, W_t, Y_t) + g_w w_t V_{t,W}(x, W_t, Y_t) \\
& - W_t c_\xi\left(\xi\right) + \xi\left(V_t\left(x + \lambda^i \bar{x}, W_t, Y_t\right) - V_t\left(x, W_t, Y_t\right)\right) \\
& - W_t c_\zeta\left(\zeta\right) + \zeta\left(p + (1-p)F(x)\right) V_t\left(x, W_t, Y_t\right) \\
& - \left(\tau\left(x\right) + \delta\right) \kappa W_t x \Bigg\}
\end{aligned}
$$

Next, guess and verify that $V_t(x, Y_t, w_t) = Y_t v(x)$, where

$$\begin{aligned}\big(\rho + \delta + \tau\,(x)\big)\, v\,(x) = \max_{\xi,\zeta\geq 0} \Big\{ & \pi\,(x) - g_x x v'\,(x) \\ & - w c_\xi\,(\xi) + \xi\Big(v\left(x + \lambda^i \bar{x}\right) - v\,(x)\Big) \\ & - w c_\zeta\,(\zeta) + \zeta\,\big(p + (1-p)F(x)\big)\, v\,(x) \\ & - \big(\tau\,(x) + \delta\big)\,\kappa w x \Big\}\end{aligned}$$

Use that $\rho = r - g_Y$, $V_{t,W}(x, W_t, Y_t) = 0$, and $w = W_t/Y_t$.

B.2 Step Size in the Quality Space

I show that there exists innovation step size functions, $\lambda^i(a, A)$ and $\lambda^e(a, A)$, in the absolute quality space $a \in \mathcal{A}$, so that the step sizes in the relative quality space $x \in \mathcal{X}$ are constant at $\lambda^i \bar{x}$ and $\lambda^e \bar{x}$, respectively.

$$\begin{aligned} x + \lambda^i \bar{x} &= \frac{\big(a + \lambda^i(a, P)\big)^\sigma}{A^{\sigma-1}} \\ \left(x A^{\sigma-1} + \lambda^i \bar{x} A^{\sigma-1}\right)^{1/\sigma} &= \left(a^\sigma + \lambda^i A^{\sigma-1}\right)^{1/\sigma} \\ &= a + \lambda^i\,(a, A) \\ \bar{x} &= 1 \end{aligned}$$

Thus, for any λ^i, there is $\lambda^i(a, A) = \left(a^\sigma + \lambda^i A^{\sigma-1}\right)^{1/\sigma} - a$ that ensures the constant step size in the relative quality space. Likewise, $\lambda^e(a, P) = \left(a^\sigma + \lambda^e A^{\sigma-1}\right)^{1/\sigma} - a$ ensures the constant step size in the relative quality space.

B.3 Derivation of the Distribution Equation

The discrete-time analog of the distribution equation is given by the following equation:

$$
\begin{aligned}
F_t(x) \approx\ & F_{t+\Delta}(x + \Delta g_x x) - \Delta\delta F_{t+\Delta}(x + \Delta g_x x) \\
& - \int_{s\in(x-\lambda^i\bar{x},\bar{x}]} \Delta\left(\xi(s) + \psi\right) dF(s) \\
& - (1-p)\left(\int_{s>x} M\Delta\left(\zeta(s) + \psi\right) dF(s) + \Delta\chi \int_{s>x-\lambda^e\bar{x}} dF(s)\right) F(s) \\
& + p\left(\int_{s\leq x} M\Delta\left(\zeta(s) + \psi\right) dF(s) + \int_{s\leq x-\lambda^e\bar{x}} \Delta\chi dF(s)\right)
\end{aligned}
$$

Divide both sides by Δ, and let $\Delta \to 0$.

B.4 Untargeted Moments

Figure B.1 compares the fractions of vertically (upper) and horizontally (lower) innovating establishments in each lagged employment decile group from the data and model. While the calibration targets the differences in the fractions of innovating establishments between the first and the second three decile groups, the model predicts far larger differences between the largest establishments with multiple product lines in the ninth and the tenth deciles and the rest of the establishments. This is because vertical and horizontal innovation events are independent across product lines. If a product line has a 1% probability of horizontal innovation per month, an establishment with one product line has a 1% probability of expanding to a new product line. On the other hand, an establishment with two product lines has approximately two percentage points of expanding to at least one product line. The size dependency is not as strong as what is suggested by this model with product-line

independence of innovation events.

Figure B.1: Model Fit - Size dependency of Innovation Frequencies

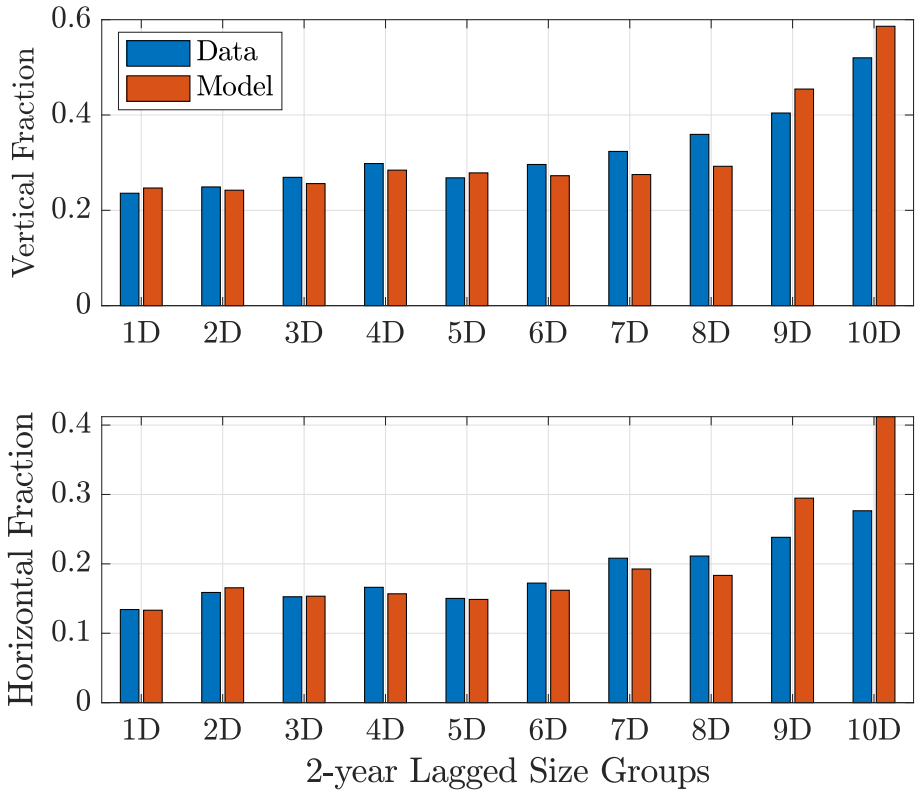


Figure B.2 shows model-generated average growth rates after vertical and horizontal innovation relative to average growth rates conditional on establishment survival. The model, as in the data, generates pre-trend among innovating establishments relative to the average establishment before innovation events because establishments that choose to innovate are the ones with better qualities. This pre-trend, however, is stronger in the data. The calibration aims to match size growth between one year after the innovation and the previous year of the innovation, but the calibrated model closely matches the size growth in the data even afterward. The model predicts size starts to grow in the innovation year, while data suggests the size grows one year after the innovation year. One potential reason is that

most establishments implement innovation in the second half of the year and the size growth after innovation is measured by the growth rate between employments as of June 30 in the innovation year and the following year.

Figure B.2: Model Fit - Relative Size Growth After Innovation Fitted Model vs. Data

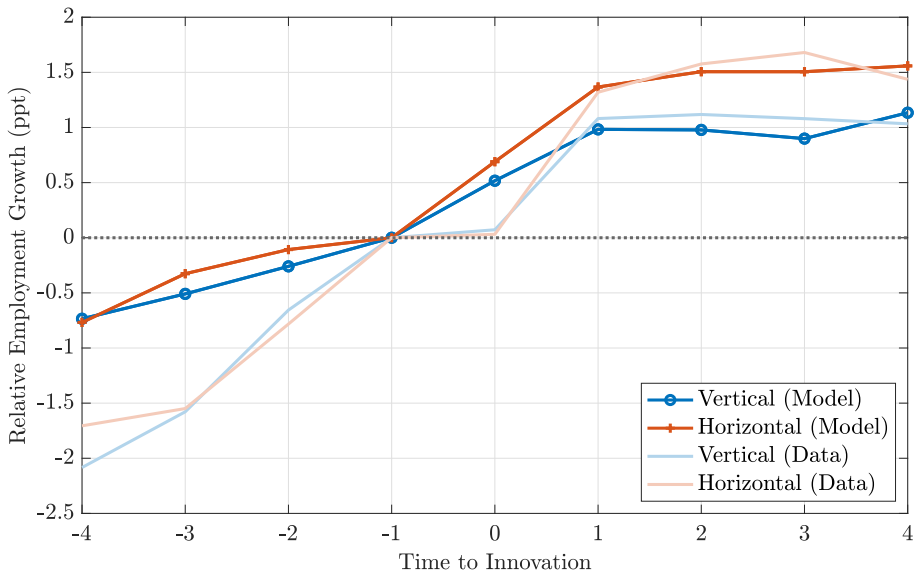


Figure B.3 compares job creation and destruction rates generated by the model to the data counterpart. The model with innovation as a single source of establishment-level shocks generates only 12% and 14% of the actual establishment-level job creation and destruction behavior. This implies that most establishment-level labor adjustments happen from demand shocks as opposed to innovation-related supply shocks as emphasized in Foster et al. (2008).

Figure B.3: Average Job Creation and Destruction Rates on the BGP

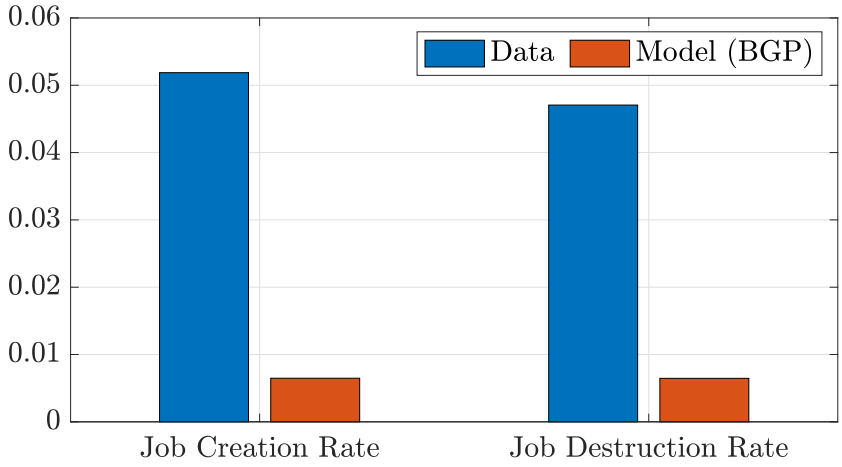
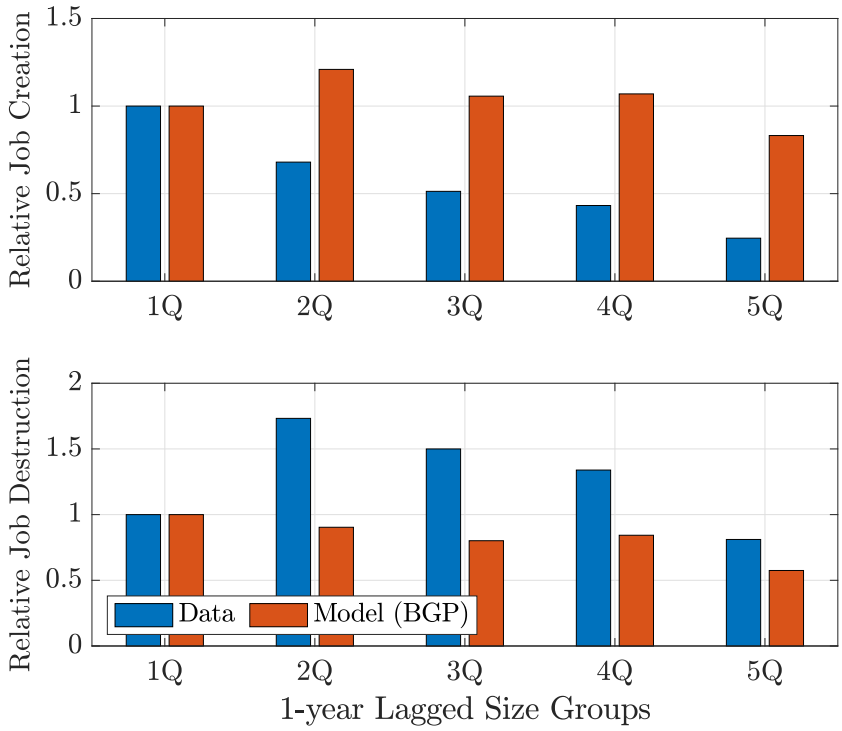


Figure B.4 expresses relative job creation and job destruction rates over lagged size quintile groups, normalized by the first quintile group of the size distribution. These job creation and destruction rates are all calculated conditional on survival. The model generates job creation and destruction rates overall decreasing in lagged size but the size-dependency is much smaller in the model than in the data. This first results from the physical constraint in the unit of workers employed in the data. The average establishment in the first quintile group has 1.5 employees in the previous year, relative to 3.5 and 5.9 employees in the second and the third quintile groups. Thus, the first group is less likely to exit the market with zero employees if it is hit by creative destruction shock or experiences severe obsolescence.

Second, in the model, most establishments up to the second quintile groups are likely to have a single product line, because expansion to a new product line is a rare event. As a result, their size difference is more likely to reflect underlying quality differences. Since high-quality firms in the second quintile group are more likely to innovate and increase their sizes than the first quintile group, the average job creation rate is higher for the second quintile group. For a similar reason, the average job destruction rate is lower for the second quintile group in the model. Nonetheless, the model can match the overall size-decreasing

property of job creation and destruction rates.

Figure B.4: Job Creation and Destruction Rates over Lagged Size Groups



C Counterfactual Appendix

C.1 Transition Path: Details

I first fix Δ, the length of time interval, to approximate the continuous-time equations. With a slight abuse of the notation, subscript t means the object evaluated at time t. Then,

the value function along the transition path satisfies the following equation:

$$
\begin{aligned}
\left(\rho+\delta+\tau_{t+\Delta}\left(x\right)\right)v_t\left(x\right) = \max_{\xi,\zeta\geq 0}\Bigg\{ & \pi_t\left(x\right)+\underbrace{\frac{v_{t+\Delta}\left(x-\Delta g_{x,t+\Delta}x\right)-v_{t+\Delta}(x)}{\Delta}}_{\approx -g_{x,t+\Delta}xv'_{t+\Delta}(x)}+\frac{v_{t+\Delta}(x)-v_t(x)}{\Delta} \\
& -w_t c_\xi\left(\xi\right)+\xi\left(v_{t+\Delta}\left(x+\lambda^i\bar{x}\right)-v_{t+\Delta}\left(x\right)\right) \\
& -w_t c_\zeta\left(\zeta\right)+\zeta\left(p_t+(1-p_t)F_{t+\Delta}(x)\right)v_{t+\Delta}\left(x\right) \\
& -\left(\tau_{t+\Delta}\left(x\right)+\delta\right)\kappa w_t x\Bigg\}
\end{aligned}
\tag{C.1}
$$

$$
\xi_t(x) = \left(\frac{v_{t+\Delta}\left(x+\lambda^i\bar{x}\right)-v_{t+\Delta}\left(x\right)\Big)}{w_t c_\xi}\right)^{\epsilon_\xi} \tag{C.2}
$$

$$
\zeta_t(x) = \left(\frac{\left(p_{t+\Delta}+(1-p_{t+\Delta})F_t(x)\right)v_{t+\Delta}(x)}{w_t c_\zeta}\right)^{\epsilon_\zeta} \tag{C.3}
$$

$$
\chi_t = \left(\frac{\int\left(p_{t+\Delta}+(1-p_{t+\Delta})F_t(x+\lambda^e\bar{x})\right)v_{t+\Delta}(x+\lambda^e\bar{x})dF_t\left(x\right)}{w_t c_\chi}\right)^{\epsilon_\chi}, \tag{C.4}
$$

In this equation system, $w_t := W_t/Y_t$ is not constant on the transition path.

The distribution along the transition path is expressed as the following equation:

$$
\begin{aligned}
F_{t+\Delta}(x) \approx & \underbrace{F_t(x+g_{x,t}x\Delta)}_{\approx\Delta f_t(x)g_x x+F_t(x)}-\Delta\delta F_t(x+g_{x,t}x\Delta) \\
& -\int_{s\in(x+g_{x,t}x\Delta-\lambda^i\bar{x},x+g_{x,t}x\Delta]}\Delta\xi_t(s)dF_t(s) \\
& -\Delta\left(1-p_{t+\Delta}\right)\left(G_{t+\Delta}(\infty)-G_{t+\Delta}(x)\right)F_t(x) \\
& +\Delta p_{t+\Delta}G_{t+\Delta}(x),
\end{aligned}
\tag{C.5}
$$

where

$$G_{t+\Delta}(x) = \int_{s \leq x} \zeta_t(s) dF_t(s) + \int_{s \leq x - \lambda^e \bar{x}} \chi_t dF_t(s) \tag{C.6}$$

$$\tau_{t+\Delta}(x) = (1 - p_{t+\Delta}) \left(\chi_t \left(\int_{s > x - \lambda^e \bar{x}} dF_t(s) \right) + \int_{s > x} \zeta_t(s) dF_t(s) \right) \tag{C.7}$$

$$\delta = p_{t+\Delta} \int \zeta_t(s) dF_t(s) + \chi_t p_{t+\Delta}. \tag{C.8}$$

The sequences of prices w_t and r_t clear the markets for labor and the goods, respectively.

$$w_t^{new} = \left(\frac{\sigma - 1}{\sigma} \right) / \left(L^s(w_t^{new}) - \int_{\Omega_t} c_\xi(\xi_{it}) di - \int_{\Omega_t} c_\zeta(\zeta_{it}) di - c_\chi(\chi_t) \right) \tag{C.9}$$

$$r_t^{new} = \frac{C_{t+\Delta} - C_t}{C_t} + \rho, \tag{C.10}$$

where $L^s(w_t^{new}) = (w_t(1 - \tau_{L,t})/\phi)^\eta$ is the labor supply curve from the first order condition of the representative household. Lastly, the depreciation rate for x can be expressed as the

following equation.

$$g_{Y,t+\Delta} = g_{A,t+\Delta} - g_{w,t+\Delta} \tag{C.11}$$

$$\begin{aligned}
g_{x.t+\Delta} &= (\rho - 1)\, g_{A,t+\Delta} = g_{Y,t+\Delta} + g_{w,t+\Delta} \\
&= \underbrace{\int \left[\int_{s \in (x-\lambda^i \bar{x}, x]} \xi_t(s) dF_t(s) \right] dx}_{=:(\rho-1) g_t^{verti}} \\
&\quad + \underbrace{p_{t+\Delta} \int \int_{s>x} \zeta_t(s) dF_t(s) dx - p_{t+\Delta} \bar{\zeta}_t}_{=:(\rho-1) g_t^{horiz(1)}} \\
&\quad + \underbrace{p_{t+\Delta} \int \chi_t \int_{s>x-\lambda^e \bar{x}} dF_t(s) dx - p_{t+\Delta} \chi}_{=:(\rho-1) g_t^{entr(1)}} \\
&\quad + \underbrace{(1 - p_{t+\Delta}) \int \left(\int_{s>x} \zeta_t(s) dF_t(s) \right) F_t(x) dx}_{=:(\rho-1) g_t^{horiz(2)}} \\
&\quad + \underbrace{(1 - p_{t+\Delta}) \int \left(\chi_t \int_{s>x-\lambda^e \bar{x}} dF_t(s) \right) F_t(x) dx}_{=:(\rho-1) g_t^{entr(2)}},
\end{aligned} \tag{C.12}$$

where $g_{Y,t+\Delta} = (C_{t+\Delta} - C_t)/C_t$ and $\bar{\zeta}_t := \int \zeta_t(s) dF_t(s)$.

The equilibrium on the transition path is defined as value function $v_t(x)$; policy function $\xi_t(x)$, $\zeta_t(x)$; product line distribution $f_t(x)$; creative destruction rates $\tau_t(x)$; depreciation and growth rates $\mu_{x,t}$ and $\mu_{Y,t}$; entry rates χ_t; and a path for wage w_t and interest rate r_t such that

1. Value function and the policy function satisfy the limit of Equation C.1-C.3 when $\Delta \to 0$.

2. Distribution function satisfies the limit of Equation C.5-C.8 when $\Delta \to 0$.

3. The price schedule satisfies the limit of Equation C.9 and C.10 when $\Delta \to 0$.

4. The depreciation and the growth rates are consistent with the limit of Equation C.11 and C.12 with $\Delta \to 0$.

5. The path of entry rate satisfies Equation C.4 with $\Delta \to 0$.

The algorithm to find the equilibrium along the transition path is as follows:

1. Fix T, a very large number. I set $T = 100$ years in my calculation. This will be where the economy converges (approximately) to the new steady state.

2. Start with an arbitrary sequence for w_t, r_t, g_t, p_t, $\tau_t(x)$ and $f_t(x)$.

3. Given the rest, update $v_t(x)$, $\xi_t(x)$, $\zeta_t(x)$, and χ_t backward from $t = T$ to $t = 0$. Notice that the value at $t = T + \Delta$ will be the value at the new steady state.

4. Given the updated policy function, update $f_t(x)$, $\tau_t(x)$, and p_t forward from $t = 0$ to $t = T$. Notice that the initial f_t at $t = 0$ is the distribution at the old steady state.

5. Given the updated policy and distribution function, update w_t, r_t, and g_t forward, using Equations C.9, C.10, and C.12, respectively. Go back to step 3 until they converge.

After the computation, we get $\{Y_t^{BS}\}_{t=0}^T$ and $\{Y_t^{TR}\}_{t=0}^T$ as our baseline BGP and transition path, respectively. Then, we can calculate the welfare of each path evaluated at time 0. Let W_0^{BS}, W_0^{TR} denote the welfare of the representative household on the baseline and the transition paths, respectively. The metrics described in Table 6 are defined by the following equations:

$$\Delta Y_0 / Y_0^{BS} := \left\{ Y_0^{tr} - Y_0^{BS} \right\} / Y_0^{BS}$$
$$t | Y_t^{TR} = Y_t^{BS} := \inf \left\{ t \geq 0 | Y_t^{TR} \geq Y_t^{BS} \right\}$$

Next, the consumption-equivalent variation (CEV) is derived by the following equations:

$$
\begin{aligned}
W_0^{TR} &= \int_0^\infty e^{-\rho t} \log\left(\left(1+CEV\right) Y_t^{BS}\right) dt \\
&= W_0^{BS} + \log\left(1+CEV\right) \int_0^\infty e^{-\rho t} dt \\
&= W_0^{BS} + \frac{\log\left(1+CEV\right)}{\rho} \\
CEV &:= \exp\left(\rho\left(W_0^{TR} - W_0^{BS}\right)\right) - 1
\end{aligned}
$$

I further decompose the change in welfare into vertical, horizontal, entrant innovation margins in the following way:

$$
\begin{aligned}
W_0 &:= \int_0^\infty e^{-\rho t} \log\left(e^{\int g_s ds} Y_0\right) dt - \int_0^\infty e^{-\rho t} \frac{\phi L_t^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} dt \\
&= \int_0^\infty e^{-\rho t} \int_0^t g_s ds dt + \int_0^\infty e^{-\rho t} \log Y_0 dt - \int_0^\infty e^{-\rho t} \frac{\phi L_t^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} dt \\
&= \int_0^\infty e^{-\rho t} \int_0^t g_s^{verti} ds dt + \int_0^\infty e^{-\rho t} \int_0^t \left(g_s^{horiz(1)} + g_s^{horiz(2)}\right) ds dt \\
&\quad + \int_0^\infty e^{-\rho t} \int_0^t \left(g_s^{entr(1)} + g_s^{entr(2)}\right) ds dt + other\, terms \\
CEV &\approx \rho\left(W_0^{TR} - W_0^{BS}\right) \\
&= \rho\left(\int_0^\infty e^{-\rho t} \int_0^t g_s^{verti,TR} ds dt - \int_0^\infty e^{-\rho t} \int_0^t g_s^{verti,BS} ds dt\right) \\
&\quad + \rho\left(\int_0^\infty e^{-\rho t} \int_0^t \left(g_s^{horiz(1),TR} + g_s^{horiz(2),TR}\right) ds dt - \int_0^\infty e^{-\rho t} \int_0^t \left(g_s^{horiz(1),BS} + g_s^{horiz(2),BS}\right) ds dt\right) \\
&\quad + \rho\left(\int_0^\infty e^{-\rho t} \int_0^t \left(g_s^{entr(1),TR} + g_s^{entr(2),TR}\right) ds dt - \int_0^\infty e^{-\rho t} \int_0^t \left(g_s^{entr(1),BS} + g_s^{entr(2),BS}\right) ds dt\right) \\
&\quad + residuals
\end{aligned}
$$

Then, I define the following metrics in Table 6, where g_0^{verti}, g_0^{horiz}, g_0^{entr} are the contributions of vertical, horizontal, and entrant innovations in welfare change, respectively:

$$\Delta g_0^{verti} := \rho \left(\int_0^\infty e^{-\rho t} \int_0^t g_s^{verti,TR} ds dt - \int_0^\infty e^{-\rho t} \int_0^t g_s^{verti,BS} ds dt \right)$$

$$\Delta g_0^{horiz} := \rho \left(\int_0^\infty e^{-\rho t} \int_0^t \left(g_s^{horiz(1),TR} + g_s^{horiz(2),TR} \right) ds dt - \int_0^\infty e^{-\rho t} \int_0^t \left(g_s^{horiz(1),BS} + g_s^{horiz(2),BS} \right) ds dt \right)$$

$$\Delta g_0^{entr} := \rho \left(\int_0^\infty e^{-\rho t} \int_0^t \left(g_s^{entr(1),TR} + g_s^{entr(2),TR} \right) ds dt - \int_0^\infty e^{-\rho t} \int_0^t \left(g_s^{entr(1),BS} + g_s^{entr(2),BS} \right) ds dt \right)$$

Lastly, the job creation and destruction rates in Table **??** are defined by the following equations. Again, $\Delta = 1$.

$$\begin{aligned}
JC_t :=& \int \Delta \xi_{t-\Delta}(x)(l_t(x + \lambda^i \bar{x}) - l_{t-\Delta}(x) - \Delta g_x x l'_{t-\Delta}(x)) dF_t(x) \\
&+ \int \Delta \zeta_{t-\Delta}(x)(p_t + (1-p_t)F_{t-\Delta}(x))(l_t(x) - \Delta g_{x,t} x l'_{t-\Delta}(x)) dF_{t-\Delta}(x) \\
&+ \Delta \chi_{t-\Delta} \int (p_t + (1-p_t)F_{t-\Delta}(x + \lambda^e \bar{x})) l_t(x + \lambda^e \bar{x}) dF_{t-\Delta}(x) \\
&+ \int (1 - \Delta \bar{\zeta}_{t-\Delta} - \Delta \bar{\xi}_{t-\Delta} - \Delta \bar{\chi}_{t-\Delta} - \Delta \bar{\tau}_t - \Delta \delta) \max\{l_t(x) - l_{t-\Delta}(x) - \Delta g_t x l'_{t-\Delta}(x), 0\} dF_{t-\Delta}(x) \\
JCR_t :=& JC_t / (L_{t-\Delta} - c_\chi(\chi_{t-\Delta})) \\
l_t(x) :=& x \frac{\sigma - 1}{w_t \sigma} + c_\xi(\xi_t(x)) + c_\zeta(\zeta_t(x)) \\
\bar{\zeta}_{t-\Delta} =& \int \zeta_{t-\Delta}(x)(p_t + (1-p_t)F_{t-\Delta}(x)) dF_{t-\Delta}(x) \\
\bar{\xi}_{t-\Delta} =& \int \xi_{t-\Delta}(x) dF_{t-\Delta}(x) \\
\bar{\chi}_{t-\Delta} =& \chi_{t-\Delta} \int (p_t + (1-p_t)F_{t-\Delta}(x)) dF_{t-\Delta}(x) \\
\bar{\tau}_t =& \int \tau_t(x) dF_{t-\Delta}(x)
\end{aligned}$$

Similarly, job destruction rates are defined by the following equations.

$$
\begin{aligned}
JD_t := & \int (\Delta\tau_t(x) + \Delta\delta) l_{t-\Delta}(x) dF_{t-\Delta}(x) \\
& + \int (1 - \Delta\bar{\zeta}_{t-\Delta} - \Delta\bar{\xi}_{t-\Delta} - \Delta\bar{\chi}_{t-\Delta} - \Delta\bar{\tau}_t - \Delta\delta) \max\{\Delta g_t x l'_{t-\Delta}(x) + l_{t-\Delta}(x) - l_t(x), 0\} dF_{t-\Delta}(x) \\
JDR_t; = & JD_t / (L_{t-\Delta} - c_\chi(\chi_{t-\Delta}))
\end{aligned}
$$

Changes in w_t imply that even non-innovating establishments may create jobs on the transition path.

C.2 Alternative Frisch Labor Supply and Entry Elasticity

I assume $\eta = 1$ and $\epsilon_\chi = 0.1$ for the baseline results in the paper. In this section, I test with alternative values for the Frisch elasticity of labor supply and establishment entry. Values for labor wedge and entry cost shifter are adjusted for each elasticity value to match the actual increase in labor supply. When $\epsilon_\chi = 0$, I assume the entry rate is fixed at the steady state, and the entrant innovation does not consume any labor.

Table C.1: Changes in Key Moments under Alternative Elasticity Values

	Baseline	$\eta = 0.3$	$\eta = 4$	$\epsilon_\chi = 0$	$\epsilon_\chi = 1$
Change in Growth (ppt)	0.07	0.07	0.07	0.07	0.13
Change in Wage	-0.098	-0.096	-0.097	-0.098	-0.091
Change in R&D Share (ppt)	-1.82	-1.80	-1.81	0.34	-2.02
Change in Job Reallocation (ppt)	-0.00	-0.00	-0.00	-0.01	0.05
Change in Vertical Innovation (ppt)	0.82	0.81	0.82	0.82	0.80
Change in Horizontal Innovation (ppt)	0.01	0.01	0.01	0.01	0.01
%Change in Job Creation at $t = 6$	345	339	342	342	320
%Change in Job Destruction at $t = 6$	-29.3	-29.3	-29.3	-29.4	-27.8
%Change in Job Creation at $t = 36$	1.3	1.3	1.3	1.0	3.6
%Change in Job Destruction at $t = 36$	-2.4	-2.2	-2.4	-2.7	1.0
%Change in Job Creation at $t = 120$	1.3	1.3	1.3	1.0	3.6
%Change in Job Destruction at $t = 120$	-0.6	-0.5	-0.6	-0.9	2.8
CEV (%)	5.88	6.03	5.74	6.50	5.93
Initial Output Change (%)	0.97	0.97	0.99	1.05	1.16
Vertical Innovation (%)	0.83	0.82	0.82	0.83	0.85
Horizontal Innovation (%)	0.05	0.05	0.05	0.03	0.18
Entrant Innovation (%)	0.10	0.10	0.10	0.05	0.49
Labor Supply (%)	-3.26	-3.01	-3.35	-2.59	-3.40

Note: The baseline is under $\eta = 1$ and $\epsilon_\chi = 0.1$. The columns that start with η assumes those values for η with $\epsilon_\chi = 0.1$, and the others starting with ϵ_χ assumes the values for ϵ_χ with $\eta = 1$. The first panel calculates changes in the steady-state moments. See Section 5.2 for the explanations. The second and the third panels summarize changes on the transition path. JCR and JDR stand for job creation and destruction rates, respectively. For the formal definitions of transition path moments, see Appendix C.1.

Table C.1 shows how aggregate moments change between the pre-reform steady state and the post-reform steady state or on the transition path from the pre-reform steady state to the post-reform steady state. All moments are relatively flat, except the consumption equivalent variation (CEV). The CEV is larger with a lower η, and these changes in CEV come entirely from the changes in disutility from labor associated with the same increase in labor supply. If I consider welfare changes only from the shape of a consumption profile, they are modestly flat or increase in Frisch elasticity because the higher elasticity implies smaller wage responses to the increase in demand after the Hartz reform. As a result, the

wage decrease is larger with a higher elasticity value, with a larger increase in welfare from more growth through vertical innovation.

The lower entry elasticity is associated with smaller incentives for vertical innovation from more frequent creative destruction. As a result, R&D labor demand decreases from a lower vertical innovation rate despite more entries, and the welfare increase is smaller. In the case of $\epsilon_\chi = 1$, the entry increase contributes to higher welfare unlike other cases, but the increment is not enough to cancel out the decrease in welfare from the reduction in vertical innovation.

C.3 R&D Tax Incentive

In this section, I compare the steady states with and without R&D subsidies. Germany began to implement a new R&D tax incentive program in 2020. After the program, 25% of R&D expenditures are exempt from corporate taxes. The average corporate tax rate is 15%, so the subsidy rate is $(0.75 \times 0.85 + 0.25 - 0.85)/0.85 = 4.41\%$. I still assume $\eta = 1$ and $\epsilon_\chi = 0.1$. I assume a government-led R&D subsidy where the labor costs associated with R&D are subsidized with the fixed rates, $s_\xi = s_\zeta = s_\chi = 4.41$ for vertical, horizontal, and entrant innovations respectively, which are subsidized by lump-sum taxes.

Table C.2: Key Changes in Steady State Moments under R&D Tax Incentive

Moment	Baseline	Vertical Subsidy	Horizontal Subsidy	Entrant Subsidy	All Subsidy
Output Growth Rate	1.30	1.33	1.31	1.30	1.33
Wage	1.000	1.001	1.001	1.005	1.007
R&D Labor Share	27.4	27.6	27.6	27.2	27.6
Labor Reallocation	1.73	1.74	1.74	1.73	1.75
Vertical Inovation	16.9	17.3	16.9	16.8	17.3
Horizontal Innovation	0.53	0.53	0.54	0.53	0.54
Job Creation from Vertical Innovation	26.2	26.6	26.2	26.1	26.6
Job Creation from Horizontal Innovation	26.8	26.7	27.1	26.8	26.9
Job Creation from Entrant Innovation	46.9	46.7	46.7	47.1	46.6
Job Destruction from Creative Destruction	13.2	13.1	13.3	13.2	13.3
Job Destruction from Obsolescence	29.0	29.4	29.1	29.1	29.5
Job Destruction from Exogenous Exit	57.8	57.5	57.6	57.7	57.2
Growth from Vertical Innovation	90.8	91.0	90.6	90.7	90.7
Growth from Horizontal Innovation	2.8	2.7	2.9	2.8	2.9
Growth from Entrant Innovation	6.4	6.3	6.5	6.5	6.5

Note: All in percent, except Wage. All flow rates are annual. The baseline has no tax incentive. The following three columns correspond to the case with only one type of vertical, horizontal, and entrant innovations subsidized. The last column corresponds to the case where all types of innovations are subsidized with equal rates. See Section 5.2 for the explanations of each row.

The key steady-state and transition path moments are shown in Table C.2. The first column corresponds to the baseline case before the tax incentives, and the last column corresponds to the full incentive case that favors all types of innovations. The three columns in the middle correspond to the case that implements the tax incentive for only vertical innovation $s_\xi > 0$, only horizontal innovation $s_\zeta > 0$, and only entrant innovation $s_\chi > 0$. The changes are small at the steady state. Nonetheless, the policy increases growth rate through higher average vertical innovation rates, as seen in the second column where the increase output growth rate is more pronounced by subsidizing vertical innovation only. When only entrant or horizontal innovations are subsidized, the steady-state growth rate decreases because the increase in these innovation types tend to decrease their average step

sizes.

Table C.3 displays statistics from the transition path analyses. In all cases, the output jumps down initially due to more labor used for incumbent and entrant innovations in exchange for faster economic growth. It takes years for the transition path to catch up with the baseline economy[33]. However, in the full reform case, such a tax incentive program actually contributes to a welfare loss because horizontal and especially entrant innovations tend to soak up an excessive amount of resources needed for production without contributing to productivity growth. When only vertical innovation is subsidized, there is a welfare gain.

Table C.3: Aggregate Moments over Transition Path under R&D Tax Incentive

Moment	Vertical Subsidy	Horizontal Subsidy	Entrant Subsidy	All Subsidy
Consumption Equivalent Variation	0.20	-0.02	-0.70	-0.57
Initial Output Change	-0.14	-0.06	-0.49	-0.70
Vertical Innovation	0.39	0.01	-0.04	0.36
Horizontal Innovation	0.00	0.02	0.00	-0.01
Entrant Innovation	0.00	0.03	0.01	-0.01
Labor Supply	-0.05	-0.02	-0.19	-0.26
Catch-up Point (years)	5.4	14.4	-	28.5

Note: The baseline has no tax incentive. The first three columns correspond to the case with only one type of vertical, horizontal, and entrant innovations subsidized. The last column corresponds to the case where all types of innovations are subsidized with equal rates. Consumption Equivalent Variation is a permanent increase in consumption for the representative household in the baseline economy to have the same lifetime utility as in the transition economy. Vertical, Horizontal, and Entrant Innovations are the contributions of each innovation on the welfare change. Labor Supply is the change in welfare from increased labor supply. Catch-up point is the time needed for the transition economy to catch up with the baseline economy without the R&D tax incentive. The rest of welfare change comes from additional production labor over the transition path. For the formal definitions, see Appendix C.1.

[33] In the case of only subsidizing entrant innovation in the third column, the transition path never catches up with the baseline BGP. This is because the higher frequency of entrant innovation reduces cross-sectional dispersion of relative qualities and step-size of horizontal and entrant innovations.

C.4 The Role of Vertical Innovation

To quantify the role of vertical innovation in the performance of the model, I remove the vertical innovation margin in the model. In this case, the economy grows only by horizontal and entrant innovation. I re-calibrate the model without vertical innovation and implement the same counterfactual exercise about the same Hartz reform. Table C.4 demonstrates the targeted moments after calibration, and Table C.5 shows the calibrated parameters without vertical innovation. The model cannot match the size growth of non-innovating establishments relative to the average establishment. The reason for this is that horizontal innovation is associated with a small increase in size. Successful horizontal innovation adds another product line that has the same productivity. In this case, conditional on successful horizontal innovation, the size growth of an establishment with a single product line rate must be 100%. However, empirically, this size increment is about 3% in an absolute term and 1.5 percentage points relative to an average establishment. This means that even among the 17.2 percent of establishments that experienced horizontal innovation, the success rate of expansion is low. Back-of-the-envelope calculation gives that, if all establishments own a single product line, the fraction of product lines added because of horizontal innovation is only about $0.17 \times 0.018 \approx 0.3$ percent per year. Moreover, entrants have smaller shares. As a result, without vertical innovation, the depreciation rate of x is low, which lowers the degree of job destruction by obsolescence and creative destruction.

Table C.4: Targeted Moments without Vertical Innovation

Moment	Data	Model	Target
Growth Rate (%)	1.3	1.3	-
Fraction of Vertical Innovation (%)	29.4	-	
Fracion of Horizontal Innovation (%)	17.3	17.2	c_ζ
Fraction of Vertical Innovation (Medium - Small, ppt)	3.6	-	-
Fraction of Horizontal Innovation (Medium - Small, ppt)	2.0	2.4	ϵ_ζ
Employment Growth (Vertical - Average, ppt)	1.2	-	-
Employment Growth (Horizontal - Average, ppt)	1.4	0.8	q^h
Employment Growth (No Innovation - Average, ppt)	-1.6	-0.4	σ
Employment Share of Establishment Age ≤ 1 (%)	1.1%	1.9%	λ^i
Establishment Share of Establishment Age ≤ 1 (%)	1.9%	2.1%	χ

Note: The Medium and the Small groups of establishments mean the first and the second tertile groups in the size distribution at the sector level after trimming establishments at the 10th size decile, respectively. All size tertile groups then are pooled across industries. All moments are constructed in the same way as the survey. Employment growth rates are measured over two years conditional on survival. Vertical-Average means the employment growth after vertically innovating establishments relative to the average survived establishments, and other relative rates are defined similarly. All observations are weighted by the cross-sectional weights from the data provider except the size growth rates. Size growth rates are weighted by the cross-sectional weight multiplied by the lagged size.

In Table C.5, a lower value for the elasticity of substitution σ is needed to increase the output growth rate relative to the quality depreciation rate. The low σ is translated to lower production labor input demand, while the demand for R&D labor remains high to match innovation rates. As a result, the share of R&D labor almost triples in Table C.6.

Table C.5: Values of Calibrated Parameters without Vertical Innovation

	Parameter	Value
λ^e	Step Size of Horizontal Innovation (%)	4.1
σ	Elasticity of Substitution across Products	1.0
c_ζ	Horizontal Innovation Cost Shifter	9.64e+06
ϵ_ζ	Horizontal Innovation Cost Elasticity	0.28
$12 \times \chi$	Annual Share of Entrant Draws (%)	1.1
q^h	Probability of Successful Expansion (%)	2.31%

Table C.6: The Steady State without Vertical Innovation

Moment	Baseline	No Vertical
Output Growth Rate	1.30	1.32
Wage	1.000	1.000
R&D Labor Share	27.4	73.1
Job Reallocation	1.73	1.32
Vertical Innovation	16.9	-
Horizontal Innovation	0.53	0.41

Note: All in percent, except Wage. All flow rates are annual. The innovation rates are defined per product line and calculated conditional on successful innovation. Thus, q^v and q^h are multiplied with vertical and horizontal innovation rates, respectively. The job reallocation rate is defined in Section 3.4.

In Table C.7, because of this small fraction of product line reallocation caused by horizontal innovation, this horizontal innovation is still less important than the entrant innovation in both job creation and in aggregate growth. Again, exogenous exits make most job destruction.

Table C.7: The Growth and Labor Reallocation Decomposition without Vertical Innovation

Moment	Baseline	No Vertical
Job Creation from Vertical Innovation	26.2	-
Job Creation from Horizontal Innov.	26.8	23.0
Job Creation from Entrant Innovation	47.0	77.0
Job Destruction from Creative Destruction	13.1	20.7
Job Destruction from Obsolescence	29.0	3.8
Job Destruction from Exogenous Exit	57.8	75.5
Growth from Vertical Innovation	90.8	-
Growth from Horizontal Innovation	2.8	16.7
Growth from Entrant Innovation	6.4	83.3

Note: The job creation and destruction are defined at the product line level. Specifically, job creation happens after vertical, horizontal, and entrant innovation. Job creation by each innovation is the frequencies of innovation multiplied by the increments in production and R&D labor hired by the innovated product line after each innovation. Job destruction happens with quality depreciation, creative destruction, and exogenous exits. Job destruction is calculated similarly by multiplying the event rate by the job loss conditional on the event. The growth rate decomposition formula is described in Section 3.7.

Tables C.8 and C.10 show results from the Hartz exercise without the vertical innovation margin in the model. I set $\kappa = 13 \times L = 2.06$ using the same logic as the baseline model. For ω_L before the reform, the labor wedge must be larger to match the actual increase in labor supply. This is because the R&D labor demand is less elastic than the production labor demand, and R&D labor share increases with a low value of σ without vertical innovation margin. Thus, I calibrate ω_L without vertical innovation to match the decrease in BGP wage after the reform in the full model; implying that, in the steady state, the reduction in innovation costs after the reform is the same in relative terms. In this case, $\omega_L = 17\%$.

The increase in the steady-state productivity growth is similar even without vertical innovation in Table C.8. Even with the small responses in horizontal and entrant innovations, the steady-state growth rate increases with a small elasticity of substitution (σ). Because of

the inelastic reaction in horizontal and entrant innovation rates, the changes in job reallocation are small. Nonetheless, unlike the baseline result, the average job reallocation rate in the steady state increases after reform.

Table C.8: The Steady State Effect of Hartz reform without Vertical Innovation

Moment	Baseline	No Vertical
Change in Output Growth Rate (ppt)	0.07	0.07
Change in Wage	-0.0981	-0.0982
Change in R&D Labor Share (ppt)	-1.8	-0.1
%Change in Avg. Job Reallocation Rate	-0.2	0.5
Change in Vertical Innovation (ppt)	0.8	-
Change in Horizontal Innovation (ppt)	0.01	0.01

Note: All in percent, except Wage. All flow rates are annual. The innovation rates are defined per product line and calculated conditional on successful innovation. Thus, q^v and q^h are multiplied with vertical and horizontal innovation rates, respectively. The job reallocation rate is defined in Section 3.4.

Table C.9 shows the instantaneous job creation and destruction rates for $\Delta = 1$ 6 months, 3 years, and 10 years after the reform. The model without vertical innovation also generates sharp increases in job creation rates relative to the pre-reform BGP. However, declines in job destruction rates are overall smaller with more creative destruction from horizontal and entrant innovations. 10 years after the reform, the model without vertical innovation predicts an increase in job destruction, while the baseline model still generates a job destruction rate lower than the pre-reform steady state.

Table C.9: The Job Turnover Effect of Hartz reform without Vertical Innovation

	Baseline	No Vertical
%Change in Job Creation at $t = 6$	345	595
%Change in Job Destruction at $t = 6$	-29.3	-3.7
%Change in Job Creation at $t = 36$	1.4	15.9
%Change in Job Destruction at $t = 36$	-8.6	-3.7
%Change in Job Creation at $t = 120$	1.3	4.2
%Change in Job Destruction at $t = 120$	-0.6	2.1

Note: Job creation and destruction rates over the transition path are defined with $\Delta = 1$ in Appendix C.1.

In Table C.10, the increase in labor supply improves welfare by 7.4%, and horizontal innovation now takes a significant fraction of the welfare increase. This is larger than the baseline increase in welfare, but the additional welfare gain comes from a smaller increase in disutility from more labor supply. Before taking the labor disutility into account, the welfare gains from more output amount to 9.14(5.87 + 3.27)% of a permanent consumption increase in the baseline case, which is higher than the 8.76(7.44+1.32)% increase in the case without vertical innovation.

These elastic responses in entrant and horizontal innovations soak up production labor in the initial period after the reform, lowering the output at $t = 0$ relative to the pre-reform BGP. In total, the contribution of the entrant and horizontal innovations to welfare is 0.86 percentage points, slightly smaller than the baseline welfare gains from productivity growth. Notably, the welfare effect of entrant innovation is negative, despite the increase in entry rates.

Horizontal and entrant innovations can have a small impact on growth than vertical innovation because of their innovation step size adjustments. Specifically, the step sizes of these innovations automatically decrease when they happen more frequently. For example, if an establishment makes a successful creative destruction shock and replicates its quality on

another product line, the dispersion in qualities between the existing and the new product line becomes zero. Because the average step-size of horizontal and entrant innovation increases with the dispersion of qualities at the product-line level, the impact of each innovation gets smaller. For horizontal innovation this adjustment is canceled out by the selection; i.e. high-quality product lines are more incentivized to expand horizontally. For entrant innovation, the direction of learning and expansion is completely random. Thus, it has a negative impact on productivity growth despite an increase in frequency.

Table C.10: The Welfare Effect of Hartz reform without Vertical Innovation

	Baseline	No Vertical
Consumption Equivalent Variation	5.87	7.44
Initial Output Change	0.97	-2.41
Vertical Innovation	0.83	-
Horizontal Innovation	0.05	1.00
Entrant Innovation	0.10	-0.14
Labor Supply	-3.27	-1.32

Note: All in percent. The Consumption Equivalent Variation is a permanent increase in consumption for the representative household in the baseline economy to have the same lifetime utility as in the transition economy. Vertical, Horizontal, and Entrant Innovations are the contributions of each innovation on the welfare change. Labor Supply is the change in welfare from increased labor supply. The rest of welfare change comes from additional production labor over the transition path. For the formal definitions, see Appendix C.1.